Reality as a Consensus Protocol: The Fixed-Point Computation That Implements Physics

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Abstract

The consensus layer of OPH models a universe as a finite network of observer patches. Patches can disagree on their overlaps. Repair moves reduce those disagreements until the public state settles. The computation is fixed-point selection of a public normal form, and cosmic history is the internal readout of that form.

The paper proves the finite mechanics used by the rest of the OPH stack: validated repair, exact descent, schedule-independent normal forms under the declared hypotheses, loop obstructions, stable records, refinement control, and a fair-block theorem for noisy consensus.

The paper's own no-go theorem (Theorem 2.6) sets the scope of the title: the bare consensus reduct does not determine a Lorentzian metric or the Einstein equation, so "implements physics" names the substrate role of the fixed-point computation, and all geometric and dynamical content enters through the separately declared branch structure of the companion papers.

What This Paper Contributes

The mathematics of repair is old in the right places. Constraint satisfaction, rewriting systems, well-founded descent, local-diamond confluence, inverse limits, and recovery channels all enter this paper with their standard meanings. OPH adds the physical reading: each variable is an observer patch, each constraint is an overlap-visible record check, and each accepted rewrite is a local repair move that must descend to the quotient seen by neighboring observers.

That turns consensus into the implementation layer of the theory. The paper proves when finite patch repair terminates, when the terminal public state is schedule-independent, how boundary data control uniqueness, and how cycle holonomy records the exact obstruction to global agreement. It also explains why a bare overlap graph is only a finite constraint code. Stronger language such as quantum error correction, min-cut distance, BFT liveness, or hardware speedup needs its own certificate.

1 Introduction

This paper writes the OPH consensus picture in the simplest concrete form. A universe is represented as a finite graph of observer patches. Each patch carries local state data, neighboring patches compare those data on overlaps, and local repair moves try to reconcile any mismatch. The central mathematical question is whether this repair dynamics converges to one shared world and how the unavoidable obstructions are encoded when it does not.

The word “dynamics” is used here in the rewriting-system sense. The repair schedule selects terminal normal forms and proves schedule independence under the stated hypotheses. Cosmic time is the internal readout of the selected observer-facing structure. A conventional simulation computes a surrogate history. OPH computes the fixed point whose internal record order is read as history. The patch-net algorithm serves as a theorem device for fixed-point selection. The fundamental description contains no global timeline on which spacetime contents are rendered.

A bare fixed point is not called an OPH simulation below. Definition 8.1 requires recovery-derived endogenous update, nontrivial quotient-readable records, overlap repair with schedule-independent normal form, elimination of a proper candidate basin in the selected boundary/sector fiber, and implementation/clock closure. Theorem 8.2 proves those clauses on the named finite branch and supplies a generic fixed-point and variational counterexample.

The resulting theorem package has two layers. The bare overlap network is only a finite constraint code: its codewords are the globally overlap-consistent states. Quantum error-correction, code-distance / min-cut resilience, exponential mixing, wall-clock BFT liveness, and hardware speedup are separate certified branches with additional data. The first core consensus layer concerns convergence: primitive recovery proposals are eligible only when they satisfy the touched-overlap local-fit contract, but a proposal becomes physical only through transactional acceptance with snapshot validation, protected-boundary preservation, and exact well-founded descent. This accepted finite relation induces a total, idempotent, boundary-preserving quotient normal-form map

$$\mathrm{Rep}_\lambda : Q \to Q.$$

Connected conflict components commit atomically through canonical union-collar payloads, so under repair completeness every initial quotient state has a unique schedule-independent normal form. The stronger claim that all interiors with the same boundary data settle to the same state requires preservation of that boundary data plus a unique consistent extension in the corresponding fiber; the layered functional carrier proves a finite multi-edge witness for this condition, and the functional selected-fiber branch gives the rooted obstruction-check form. The second concerns obstructions: pairwise overlap agreement does not ensure a global solution, and the obstruction is holonomic. On the abelian branch it is the cycle sum of edge data; on the genuinely noncentral branch it is a crossed-module Čech class.

Gauge symmetry enters as invariance under changes of hidden local representation that preserve overlap data. When the repair step is read only on that overlap-invariant quotient, the normal-form map descends to the gauge quotient, so physical uniqueness is a quotient statement. On the quantum lift, the same quotient-local carrier determines a unique terminal state on every declared physical observable algebra, even when microscopic representative lifts differ by gauge or sector relabelings inside one quotient-local glued state. The observation layer is carried by finite observer-accessible record algebras generated by central or quantitatively stable approximately commuting projectors. These results form the patch-net formulation used in the broader OPH literature, including the fixed-cutoff Bell/CHSH package on the companion microphysics surface.

This paper proves a constraint-code firewall and nine core consensus results:

1. **Constraint-code firewall.** A finite overlap net defines a finite constraint code: its codewords are exactly the globally consistent states, and $C = \Phi^{-1}(0)$ (Proposition 2.3). This is the default meaning of “the overlap network is a code.” It is not, by itself, a quantum error-correcting code and does not imply a graph min-cut formula for distance (Theorem C.10). The same firewall blocks semantic promotion by relabeling: a finite constraint, archive, repair spectrum, or reconstruction threshold remains a finite diagnostic until a source-separated

physical bridge supplies the relevant readout, residual ledger, controls, and frozen validation target.

2. **Asynchronous confluence.** For the declared accepted repair law, primitive recovery proposals pass through transactional snapshot/read/write validation, boundary/sector preservation, and exact descent. The accepted relation induces a total local quotient repair map $\mathrm{locRep}_\lambda : Q \to Q$ and a total idempotent global repair map $\mathrm{Rep}_\lambda = \overline{\mathrm{nf}}_\lambda : Q \to Q$. Atomic conflict-component commits give the local diamond on the physical quotient; under repair completeness, every fixed initial quotient state has a unique normal form, independent of update schedule (Proposition 3.7, Theorems 3.9 and 3.22).

3. **Cycle obstruction.** For affine overlap constraints over an abelian group, global consistency holds if and only if the holonomy vanishes on every cycle (Theorem 4.1). The parity triangle gives the minimal frustrated example, and Theorem 4.4 extends the same logic to the crossed-module higher-gauge defect hierarchy used later in the framework.

4. **Gauge quotient, selected fibers, and observable-level confluence.** When local repair is induced on the overlap-invariant quotient, the normal-form map descends to that quotient, $\mathrm{Rep}_\lambda \circ q$ is invariant under gauge/implementation hiding, and the induced terminal state on every declared physical observable algebra is unique there even when microscopic representatives differ by gauge or sector relabelings inside one quotient-local glued state. If a boundary/sector map is preserved and each consistent boundary fiber has at most one quotient extension, then all initial states with that boundary value settle to the same quotient normal form; the layered functional carrier proves $H_B \wedge H_{\mathrm{fib}}$ on a finite multi-edge, multi-step carrier, while the functional selected-fiber branch proves a nontrivial rooted case where multiple same-boundary interiors exist, inconsistent candidates are eliminated, and surviving candidates share one quotient normal form (Theorems 5.2, 5.4, 5.7, 5.9, Corollaries 5.3, 5.8, 5.11, and Theorem 5.17).

5. **Refinement-limit consensus classes.** On a separated cofinal refinement system whose restriction maps commute with the finite-stage normal-form and holonomy maps, the quotient normal forms and holonomy obstructions assemble into unique inverse-limit classes with finite-stage visibility (Theorem 6.2).

6. **Coarse-graining compatibility.** On any refinement system whose coarse-graining maps shadow finite-stage normal forms and holonomy maps with declared errors, reconciling first and then coarse-graining gives the same macroscopic law data as coarse-graining first and then reconciling, up to those errors; in the exact natural case the error is zero (Theorem 6.9).

7. **Record algebra and stability.** On the fixed-cutoff observer-accessible surface, central record projectors carry Born/Lüders measurement directly, and approximate record projectors inherit explicit $(\varepsilon, \delta_{\mathrm{rec}})$ stability bounds on the same event surface (Theorem 7.2).

8. **OPH simulation firewall.** On a selected boundary/sector fiber, "simulation" requires recovery-derived endogenous update, nontrivial quotient-readable records, strict overlap-repair descent with a schedule-independent normal form, collapse of a proper candidate basin to one consistent quotient state, and implementation/clock closure. Under the stated selected-fiber and record hypotheses, the finite OPH packet has this certificate; a generic identity fixed point or constant variational functional does not (Definition 8.1, Theorem 8.2, and Remark 8.3).

9. **Distributed one-universe realization.** A worker implementation is a presentation of one finite OPH universe only when the run starts from one global carrier and every distributed event projects to a legal monolithic repair path, a physical stutter, or a certified rollback to an earlier committed projection. Under those hypotheses, partition, worker count, schedule, restart history, and repartition metadata do not change quotient observables or observer readouts that factor through the monolithic normal form (Theorem 9.5 and Corollary 9.6).

10. **Conditional noisy fair-block consensus.** If a noisy asynchronous implementation admits fair repair blocks, a uniform expected contraction toward the exact quotient normal-form set, and controlled within-block excursions, then all long runs stay in a controlled expected tube around that exact normal-form set. In singleton boundary/sector fibers, Lipschitz observer readouts are approximately schedule-independent, and a finite Markov-kernel certificate checks the block contraction on finite exported nets (Theorem 11.1, Corollaries 11.2–11.3, and Proposition 11.4).

The repair step itself is a concrete recovery move, not a free rewrite primitive. On the fixed-cutoff collar branch, a local update is obtained from exact Markov splice or from a declared Petz/Fawzi–Renner recovery channel and then read on overlap-invariant physical data. The fixed-point theorem below isolates the separate branch conditions cleanly: the declared repair law includes the touched-overlap local-fit contract for primitive proposals, while transactional acceptance validates snapshots, protected data, unchanged registers, and exact descent. The fixed-cutoff gluing package supplies the parenthesization-independent union-collar payload used for atomic conflict-component commits. What sits above that step is repair completeness and, on the Petz branch, the support/CPTP clause stated later in Proposition C.5. A nontrivial exported rooted-tree packet domain where these clauses are proved is recorded in Definition 3.13 and Theorem 3.14; the separate layered functional carrier records the finite multi-edge boundary-reconstruction witness.

We also define a fitness functional over a finite candidate space of reconciliation laws and prove that replicator dynamics monotonically increases mean fitness (Theorem 12.4). This gives a clean mathematical model for finite-candidate law selection, not a universality theorem or a literal cosmological dynamics claim.

The results here are exact theorems about a computational model. The companion OPH manuscript uses separate support levels for structural theorems, scaling limits, quantitative particle outputs, and phenomenological continuations [2].

2 Patch Nets, Overlaps, and Global Consistency

Definition 2.1 (Patch net)**.** *Let $G = (V, E)$ be a finite connected graph. Each vertex $i \in V$ is an* ***observer patch*** *with finite local state space S_i. The global state space is*

$$\Sigma := \prod_{i \in V} S_i.$$

For each edge $e = \{i, j\} \in E$, let I_e be an interface alphabet and let

$$\pi_{i,e} : S_i \to I_e, \qquad \pi_{j,e} : S_j \to I_e$$

be the interface projection maps. A global state $s = (s_i)_{i \in V} \in \Sigma$ is ***consistent on edge*** *$e = \{i, j\}$ iff*

$$\pi_{i,e}(s_i) = \pi_{j,e}(s_j).$$

The global consistency set is

$$C := \{s \in \Sigma : \forall\, e = \{i,j\} \in E,\ \pi_{i,e}(s_i) = \pi_{j,e}(s_j)\}.$$

For exposition we use a finite pairwise-overlap graph. The hypergraph version is straightforward: replace edges by hyperedges and pairwise equality by a common interface label on each hyperedge. Nothing in the proofs depends on the pairwise restriction.

The picture: each observer holds a local state, and neighboring observers share an interface through which they can compare notes. A universe-state is physically admissible exactly when all neighbors agree on their shared data. This is a constraint satisfaction problem (CSP), and the consistent states are the codewords.

Definition 2.2 (Inconsistency potential)**.** *For each edge e, choose a weight $w_e > 0$ and a function $d_e : I_e \times I_e \to \mathbb{R}_{\geq 0}$ with $d_e(a,b) = 0 \iff a = b$. On the declared fixed-cutoff branch, d_e is the overlap score used by the local acceptance contract on that interface. Define*

$$\Phi(s) := \sum_{e=\{i,j\}\in E} w_e\, d_e\big(\pi_{i,e}(s_i), \pi_{j,e}(s_j)\big).$$

Then $s \in C \iff \Phi(s) = 0$.

Proposition 2.3 (Bare overlap nets are finite constraint codes)**.** *For every finite patch net of Definition 2.1, the consistency set $C \subseteq \Sigma$ is a finite constraint code whose codewords are exactly the globally overlap-consistent states. With the mismatch potential of Definition 2.2,*

$$C = \Phi^{-1}(0).$$

This proposition is the theorem-grade content of the unqualified phrase "the overlap network is a code." It supplies a finite constraint code, not a quantum error-correcting code, not a topological code, not a graph-theoretic formula for code distance, and not a Lorentzian or Einstein-geometry theorem.

Proof. Finiteness follows from $\Sigma = \prod_i S_i$ with finite S_i. The definition of C is a finite family of interface-equality constraints, so C is the set of satisfying assignments. Since each $w_e > 0$ and $d_e(a,b) = 0$ exactly when $a = b$, every term in Φ is nonnegative and vanishes exactly on a satisfied edge constraint. Therefore all edge constraints hold if and only if $\Phi(s) = 0$. □

Remark 2.4 (Bare consensus does not supply the cap-normal H^3 chart). Bare finite consensus supplies quotient normal forms, boundary data, and obstruction classes. It does not itself supply the future null cone, a round-cap support chart, conformal transport, a time orientation, or the H^3 observer-frame homogeneous space. The cap-normal theorem in the compact paper applies only after the separate geometric-readout/BW branch has emitted an oriented conformal S^2, non-degenerate oriented round caps, and proper-orthochronous conformal transport. This is why the finite-consensus to Einstein branch-entry arrow remains a separate proof burden.

Definition 2.5 (Bare finite consensus reduct)**.** *A bare finite consensus reduct at regulator r is*

$$\mathsf{Cons}_r = (\Sigma_r, \Gamma_r, Q_r, \Phi_r, \to_r, n_r, C_r),$$

where Σ_r is the finite presentation space, Γ_r is the presentation-redundancy groupoid, $Q_r = \Sigma_r/\Gamma_r$ is the physical quotient, Φ_r is the mismatch functional, $\to_r$ is the accepted repair relation, n_r is the quotient normal-form map, and

$$C_r = \Phi_r^{-1}(0)$$

is the globally overlap-consistent set.

Theorem 2.6 (Bare finite consensus is not Einstein-complete)**.** *The bare finite consensus reduct* Cons_r *does not determine a Lorentzian metric* g_{ab}*, stress tensor* T_{ab}*, Newton coupling* G*, area/edge operator* L_C*, generalized entropy* S_{gen}*, modular geometric flow, or the equation*

$$G_{ab} + \Lambda g_{ab} = 8\pi G\, T_{ab}.$$

Consequently, Einstein geometry is not a theorem of the bare finite consensus language.

Proof. The symbols of Cons_r describe finite states, quotienting, mismatch, accepted repair, normal forms, and consistency. They do not include a metric, curvature tensor, stress tensor, cap modular automorphism, area operator, or entropy-area normalization. Hence two model extensions can share the same Cons_r while assigning different geometry/stress data. One extension may attach Minkowski metric, $T_{ab} = 0$, and $\Lambda = 0$, so Einstein holds. Another may attach a Lorentzian metric g'_{ab} and stress tensor T'_{ab} for which

$$G'_{ab} + \Lambda' g'_{ab} \neq 8\pi G' T'_{ab}$$

somewhere. All bare consensus statements have the same truth value in the two extensions, while the Einstein equation has different truth values. Therefore the Einstein equation is not entailed by the bare reduct. □

$$\boxed{\mathsf{Cons}_r \Rightarrow \text{quotient normal forms only.}}$$

The gravity theorem used elsewhere in OPH has a separate five-axiom recovered-core dependency spine:

$$\boxed{\begin{array}{c}\mathsf{RecoveredCore}_{5\mathrm{ax}} + \mathsf{GeomRead} + \mathrm{BW}^{\mathrm{sv}}_{S^2} + \mathsf{NullStress} \\ +\mathsf{BoundedInterval} + \mathsf{FixedCapStat} + \mathsf{SmallBallArea} \\ +\mathsf{RemainderControl} + \mathsf{TimelikeCoverage} + \mathsf{TensorUpgrade} \Rightarrow \mathsf{Einstein}.\end{array}}$$

That branch is carried by the compact SM/GR and microphysics papers, not by the bare constraint-code theorem above.

So Φ is the total disagreement energy of the universe. Consistent states have zero energy. Everything else is frustrated.

3 Asynchronous Reconciliation and the Main Fixed-Point Theorem

Definition 3.1 (Recovery-derived local repair law)**.** *Fix for each patch* i *a finite collar chart* A_i*–*B_i*–*D_i *around the overlaps touched by* i*, together with a fixed local decoder from repaired collar data back to the finite patch label at* i*. A* ***law*** λ *is a family of local repair maps*

$$T_i^\lambda : \Sigma \to \Sigma \qquad (i \in V)$$

such that T_i^λ *changes only the state of patch* i *(or, more generally, only a bounded neighborhood of* i*), and the local update is induced by one of the declared OPH recovery moves on that collar, where* $\omega_{A_iB_i}(s)$ *denotes the* $A_i \cup B_i$ *marginal of the input collar state encoded by* s*:*

(i) exact Markov splice on the collar, using Theorem A.1 when $I(A_i : D_i \mid B_i) = 0$*; or*

(ii) *a declared recoverability channel*

$$(\mathrm{id}_{A_i} \otimes \mathcal{R}_i)(\omega_{A_i B_i}(s)), \qquad \mathcal{R}_i = \mathcal{R}_{\sigma_i, \mathcal{N}_i},$$

with $\mathcal{R}_i$ in the Petz/Fawzi–Renner class of Definition C.3 and Theorem A.1.

The decoder back to S_i is bookkeeping for the finite patch presentation; the physical content is the repaired collar state on the declared fixed-cutoff branch. Write $s \rightsquigarrow_i t$ iff $t = T_i^{\lambda}(s) \neq s$. These are primitive proposals awaiting acceptance as physical rewrite steps. A proposal is committed only through the transactional acceptance layer below.

Definition 3.2 (Touched-overlap potential and accepted local-fit contract)**.** *For each repair site i, let*

$$E_i^{\mathrm{touch}} := \{e \in E : \textit{the interface data on } e \textit{ may change under } T_i^{\lambda}\}.$$

Define the touched-overlap potential

$$\Phi_i(s) := \sum_{e=\{u,v\} \in E_i^{\mathrm{touch}}} w_e\, d_e\big(\pi_{u,e}(s_u), \pi_{v,e}(s_v)\big).$$

On the declared fixed-cutoff branch, a recovery-derived primitive candidate is eligible for transactional commitment only if it strictly lowers this touched-overlap score:

$$s \rightsquigarrow_i t \implies \Phi_i(t) < \Phi_i(s).$$

This is the patch-net form of the regulator-side monotone local-fit contract carried by the declared repair package.

Definition 3.3 (Overlap-associative union-collar gluing)**.** *Fix $s \in \Sigma$ and two enabled primitive proposals $s \rightsquigarrow_i t$, $s \rightsquigarrow_j u$. Write*

$$E_{ij}^{\mathrm{touch}} := E_i^{\mathrm{touch}} \cup E_j^{\mathrm{touch}}.$$

The declared repair branch is ***overlap-associative*** *if the following hold.*

(i) *If $E_i^{\mathrm{touch}} \cap E_j^{\mathrm{touch}} = \varnothing$, the two local proposals have disjoint support on the declared branch and therefore commute if they are committed as separate transactions.*

(ii) *If $E_i^{\mathrm{touch}} \cap E_j^{\mathrm{touch}} \neq \varnothing$, there is a finite union collar U_{ij} covering the interfaces in E_{ij}^{touch} such that the physical glued state on U_{ij} is parenthesization-independent on the quotient, in the sense of Proposition B.8, and the local decoders/lifts of Definition 3.1 are restriction-compatible on nested collars.*

This is the concrete compatibility package used below to build canonical aggregate payloads for conflicting repair components.

Definition 3.4 (Transactional acceptance layer)**.** *Let X denote the finite physical presentation on which a repair step is being checked: before quotienting one may take $X = \Sigma$, and on the physical branch X is the quotient by hidden representatives. Registers are the finite patch, interface, sector, and record coordinates. Fix a boundary/sector/holonomy record map*

$$B : X \to \mathcal{B}$$

and a well-founded exact measure

$$\mu : X \to (W, \prec),$$

for example a lexicographic integer vector $(N_{\mathrm{hard}}, \Phi, N_{\mathrm{unresolved}})$.

A prepared transaction is a tuple

$$\tau = (R_\tau, W_\tau, \sigma_\tau, p_\tau)$$

with read set, write set, read snapshot, and payload. It commits at $x \in X$ *only if all of the following hold:*

(i) *the snapshot is current:* $x|_{R_\tau} = \sigma_\tau$*;*

(ii) *the payload changes no register outside* W_τ*;*

(iii) *boundary, sector, holonomy, and protected record data are preserved:* $B(\mathrm{Apply}_\tau(x)) = B(x)$*;*

(iv) *exact descent holds:*

$$\mu(\mathrm{Apply}_\tau(x)) \prec \mu(x).$$

A stale, aborted, ambiguous, or obstructed transaction is not a rewrite step. The read set is ***validation-complete*** *for* μ*: if a term of* μ *can change when a register in* W_τ *changes, the whole support of that term lies in* R_τ*. For seam potentials this means reading both endpoints of every seam whose score can change.*

At a state x*, form the conflict graph of enabled primitive repair proposals, with*

$$\tau \# \sigma \quad \Longleftrightarrow \quad W_\tau \cap (R_\sigma \cup W_\sigma) \neq \varnothing \text{ or } W_\sigma \cap (R_\tau \cup W_\tau) \neq \varnothing.$$

Each connected conflict component K *is replaced by exactly one canonical aggregate transaction* τ_K*, computed on the union-collar or conflict-component support. Primitive members of* K *never commit separately. Write* $x \to y$ *for a successful aggregate transaction commit, and let* $\to^*$ *be its reflexive-transitive closure. A state is a* ***normal form*** *when no aggregate transaction is enabled.*

3.1 The quotient repair operator

The transactional layer above defines the physically accepted one-step relation. The object used as law is the induced finite normal-form map on the physical quotient, not a hidden-representative rewrite.

Definition 3.5 (Finite quotient repair presentation)**.** *A* ***finite quotient repair presentation*** *is a tuple*

$$\mathcal{P} = (\Sigma, \Gamma, q, Q, C_Q, B, \mu, \mathsf{A}, \prec_{\mathsf{A}})$$

where

$$Q = \Sigma/\Gamma, \qquad q : \Sigma \to Q$$

is the physical quotient by hidden representative data, $C_Q \subseteq Q$ *is the quotient-level consistency set,*

$$B : Q \to \mathcal{B}$$

is the protected boundary, sector, root-packet, charge, or holonomy-sector map, and

$$\mu : Q \to (W, \prec)$$

is a well-founded exact descent measure whose image on Q is finite. The finite set A consists of accepted aggregate repair transactions, equipped with a fixed total order $\prec_{\mathsf{A}}$. Each $a \in \mathsf{A}$ has a domain $D_a \subseteq Q$ and a map

$$a : D_a \to Q.$$

The accepted one-step relation is

$$x \to_{\mathcal{P}} y \quad\iff\quad \exists a \in \mathsf{A},\ x \in D_a,\ y = a(x).$$

The presentation is ***OPH-admissible*** *when:*

$$(H_B) \qquad x \to_{\mathcal{P}} y \implies B(y) = B(x),$$

$$(H_{\downarrow}) \qquad x \to_{\mathcal{P}} y \implies \mu(y) \prec \mu(x),$$

$$(H_{\diamond}) \qquad \to_{\mathcal{P}} \textit{ is locally confluent on } Q,$$

and

$$(H_{\mathrm{comp}}) \qquad x \in C_Q \quad\iff\quad \textit{no accepted aggregate transaction is enabled at } x.$$

Definition 3.6 (Local quotient repair operator)**.** *For $x \in Q$, let*

$$\mathsf{A}(x) := \{\, a \in \mathsf{A} : x \in D_a \,\}$$

be the enabled aggregate transaction set. Define

$$\mathrm{locRep}_{\lambda}(x) := \begin{cases} a_{\min}(x), & \mathsf{A}(x) \neq \varnothing, \\ x, & \mathsf{A}(x) = \varnothing, \end{cases}$$

where $a_{\min}$ is the $\prec_{\mathsf{A}}$-least enabled aggregate transaction.

Proposition 3.7 (Local quotient repair is total, protected, and descending)**.** *For every OPH-admissible finite quotient repair presentation,*

$$\mathrm{locRep}_{\lambda} : Q \to Q$$

is a total map. For every $x \in Q$,

$$B(\mathrm{locRep}_{\lambda}(x)) = B(x),$$

and either $\mathrm{locRep}_{\lambda}(x) = x$ or

$$\mu(\mathrm{locRep}_{\lambda}(x)) \prec \mu(x).$$

Finally,

$$\mathrm{locRep}_{\lambda}(x) = x \quad\iff\quad x \in C_Q.$$

Proof. The set A is finite and totally ordered, so a least enabled transaction exists whenever $\mathsf{A}(x) \neq \varnothing$. If no transaction is enabled, the definition returns x. Hence $\mathrm{locRep}_{\lambda}$ is total.

If no transaction is enabled, boundary preservation is immediate. If $a_{\min}$ is enabled, (H_B) gives $B(a_{\min}(x)) = B(x)$. The descent statement is immediate from $(H_{\downarrow})$. Since strict descent excludes $a_{\min}(x) = x$, the local operator fixes exactly those states with no enabled aggregate transaction. By (H_{comp}), these are exactly the elements of C_Q. ☐

Definition 3.8 (Global quotient repair operator)**.** *For $x \in Q$, define the canonical local-repair iterates by*

$$x_0 = x, \qquad x_{n+1} = \mathrm{locRep}_\lambda(x_n).$$

By Proposition 3.7, the sequence either stops or strictly descends inside the finite value set $\mu(Q)$. Hence there is a least $N(x)$ such that

$$x_{N(x)+1} = x_{N(x)}.$$

The ***global quotient repair operator*** *is*

$$\mathrm{Rep}_\lambda(x) := x_{N(x)}.$$

Theorem 3.9 (Global Repair is the finite quotient normal-form map)**.** *For every OPH-admissible finite quotient repair presentation,*

$$\mathrm{Rep}_\lambda : Q \to Q$$

is total and satisfies

$$\mathrm{Rep}_\lambda(x) \in C_Q, \qquad B(\mathrm{Rep}_\lambda(x)) = B(x),$$
$$\mathrm{Rep}_\lambda(\mathrm{Rep}_\lambda(x)) = \mathrm{Rep}_\lambda(x),$$

and

$$\mathrm{Rep}_\lambda(x) = x \quad \Longleftrightarrow \quad x \in C_Q.$$

If $x \to_{\mathcal{P}}^* y$ and y is terminal, then

$$y = \mathrm{Rep}_\lambda(x).$$

Thus

$$\mathrm{Rep}_\lambda = \overline{\mathrm{nf}}_\lambda : Q \to Q$$

is independent of the accepted asynchronous repair schedule.

Proof. Totality follows from finite descent. If $x_{n+1} \neq x_n$, then Proposition 3.7 gives

$$\mu(x_{n+1}) \prec \mu(x_n).$$

Since $\mu(Q)$ is finite and well founded, no infinite strictly descending sequence exists. Thus the canonical iteration reaches a fixed point x_N. By Proposition 3.7,

$$x_N = \mathrm{locRep}_\lambda(x_N) \quad \Longleftrightarrow \quad x_N \in C_Q,$$

so $\mathrm{Rep}_\lambda(x) \in C_Q$. Boundary preservation follows by induction from (H_B), hence $B(\mathrm{Rep}_\lambda(x)) = B(x)$. Idempotence follows because every point of C_Q is fixed by Proposition 3.7.

For schedule independence, $(H_\downarrow)$ gives termination of $\to_{\mathcal{P}}$, and $(H_\diamond)$ gives local confluence. Newman's lemma gives confluence. A terminating confluent rewrite system has a unique terminal normal form reachable from each initial state. The canonical iteration defining Rep_λ is one accepted repair execution, so it reaches that unique terminal normal form. Therefore every other maximal accepted repair execution from x reaches the same value. $\square$

Closure item	**Paper object**	**Result**
physical one-step repair	$\mathrm{locRep}_\lambda : Q \to Q$	Proposition 3.7
physical global repair	$\mathrm{Rep}_\lambda = \overline{\mathrm{nf}}_\lambda$	Theorem 3.9
protected data	boundary/sector map B preserved by accepted transactions	Theorem 3.9
quotient normal form	terminal point in C_Q, idempotent and schedule-independent	Theorem 3.9

Proposition 3.10 (Validation support for local mismatch measures)**.** *Suppose the exact repair measure has finite local supports,*

$$\mu(x) = \sum_{a \in A} \mu_a(x|_{S_a}).$$

If a transaction writes W, then only terms with $S_a \cap W \neq \varnothing$ can change. Consequently, the descent validation is snapshot-local once the read set contains

$$R^\mu(W) := \bigcup_{a : S_a \cap W \neq \varnothing} S_a.$$

For the OPH overlap potential

$$\Phi(x) = \sum_{e=\{i,j\}} w_e \, d_e\big(\pi_{i,e}(x_i), \pi_{j,e}(x_j)\big),$$

this means reading both endpoints of every overlap whose score may change when a written register changes.

Proof. If $S_a \cap W = \varnothing$, the transaction leaves every register read by μ_a unchanged, so that term cancels between the pre- and post-state. Every remaining term is determined by the payload together with the restriction to $R^\mu(W)$. The displayed edge-potential formula has one two-endpoint support for each overlap term, giving the stated seam rule. □

Remark 3.11 (Inputs and branch conditions). The repair step is therefore not an abstract rewrite primitive. Its declared inputs are the fixed-cutoff collar chart, either exact Markov splice or a chosen Petz/Fawzi–Renner recovery channel, a local decoder/lift back to the finite patch presentation, and the touched-overlap local-fit contract of Definition 3.2, together with the support-local disjoint-commutation clause and the restriction-compatible union-collar package of Definition 3.3. The parenthesization-independent quotient-local glue used there is supplied by Proposition B.8 from the fixed-cutoff center-sector / higher-gauge gluing package. The actual accepted step is the validated aggregate transaction of Definition 3.4. The theorem package takes repair completeness as an explicit branch condition. On the Petz branch, full CPTP action on all inputs also requires the support clause recorded in Proposition C.5. On broader branches one must also prove that the declared union-collar compatibility is preserved under refinement or branch change.

The theorem package separates the imported repair-law data from the theorem-local inputs cleanly:

Assumption 3.12 (Repair completeness)**.** *For the accepted transactional relation $\to$, $s \in C$ if and only if no aggregate repair transaction is enabled at s.*

Normal forms are exactly the globally consistent states. The dynamics is neither too weak (missing some inconsistencies) nor too strong (repairing things that were fine).

Definition 3.13 (Verified rooted-tree packet-net domain)**.** *Fix a finite rooted tree $T = (V, E, r)$. For each non-root vertex i, write $p(i)$ for its parent and write $w_i > 0$ for the weight of the edge $\{p(i), i\}$. Let A be a finite packet alphabet with $|A| \geq 2$, and let K_i be a finite hidden-label set. The patch state space is*

$$S_i = A \times K_i, \qquad s_i = (x_i, k_i).$$

For every edge $e = \{i, j\}$, the interface alphabet is $I_e = A$, and both endpoint projections read the packet component:

$$\pi_{i,e}(x_i, k_i) = x_i, \qquad \pi_{j,e}(x_j, k_j) = x_j.$$

Thus e is consistent exactly when $x_i = x_j$. Choose the weights so that

$$w_i > \sum_{j:p(j)=i} w_j \qquad \text{for every non-root } i,$$

with an empty sum equal to 0*. Define*

$$\Phi(s) = \sum_{\{p(i),i\}\in E} w_i\,\mathbf{1}[x_i \neq x_{p(i)}].$$

The repair map at a non-root vertex i is

$$T_i(s)_i = (x_{p(i)}, k_i),$$

with all other vertices unchanged; if $x_i = x_{p(i)}$, the map is a no-op. The root repair map is the identity. The hidden labels k_i are acted on by arbitrary finite gauge relabelings and are not read by any interface projection.

Theorem 3.14 (Rooted-tree packet repair completeness and quotient closure)**.** *On the domain of Definition 3.13, Assumption 3.12 is a theorem. Every enabled repair strictly decreases Φ. Every maximal asynchronous repair run terminates at the unique state*

$$x_i = x_r \quad \text{for all } i \in V,$$

with all hidden labels k_i unchanged. The normal-form map descends to the quotient by hidden gauge relabeling. If several tree repairs lie in one conflict component, the canonical aggregate transaction applies them in increasing tree depth and is the same as the corresponding serial tree repair path. The four-vertex instance with edges $r-a$, $a-b$, $a-c$, alphabet $A = \mathbb{Z}_3$, hidden labels $K_i = \mathbb{Z}_2$, and weights 5, 1, 1 *is exported as* `code/consensus/runs/verified_tree_packet_net_domain.json`*.*

Proof. First, repair completeness is immediate from the rooted tree. If $s \in C$, every edge has $x_i = x_{p(i)}$, so every non-root repair is a no-op. Conversely, if every repair is a no-op, then $x_i = x_{p(i)}$ for every non-root vertex, hence every edge is consistent and $s \in C$.

Let $i \neq r$ be enabled. Only the parent edge $\{p(i), i\}$ and child edges $\{i, j\}$ with $p(j) = i$ can change their contribution to Φ. The parent edge changes from inconsistent to consistent, contributing $-w_i$. Each child edge can increase by at most its weight. Therefore

$$\Phi(T_i(s)) - \Phi(s) \leq -w_i + \sum_{j:p(j)=i} w_j < 0.$$

So every enabled repair is accepted by the Lyapunov contract.

Since Φ takes finitely many values, every maximal repair run terminates. At a terminal state no non-root vertex differs from its parent, so the terminal packet label is x_r on every vertex. The root label and every hidden label are invariant under all repairs, so the terminal state is unique and independent of update order. Because interface projections, enabledness, Φ, and the repair maps depend only on x_i, arbitrary relabelings of the hidden K_i commute with quotienting:

$$q \circ T_i = \overline{T}_i \circ q.$$

Thus the normal-form map descends to the hidden-label quotient. $\square$

Corollary 3.15 (Physical-law map on the verified packet domain)**.** *On the rooted-tree packet domain, every gauge-invariant observable*

$$M : \prod_i (A \times K_i) \to Y$$

has a schedule-independent repaired value

$$M(\mathrm{nf}(s)),$$

and this value depends only on the quotient class of s. Hence the promoted normal-form map is usable as physical law on this verified packet branch without adding a representative-level gauge-covariance assumption.

Proof. Theorem 3.14 gives a unique terminal state for every asynchronous repair schedule and shows that the normal-form map descends to the quotient by hidden-label relabeling. A gauge-invariant observable factors through that quotient, so its value on the terminal state is independent of both the repair schedule and the hidden representative. □

Proposition 3.16 (Classical full-support Petz packet domain)**.** *Let B and D be finite packet alphabets, let the collar algebras be diagonal, and let $\mathcal{N} : \mathbb{C}^{B\times D} \to \mathbb{C}^B$ be the marginal channel $(\mathcal{N}p)(b) = \sum_d p(b,d)$. Fix a reference state σ_{BD} with $\sigma_B(b) > 0$ for every b. Let*

$$\gamma_\sigma := \min_{b\in B} \sigma_B(b) > 0.$$

Define the Petz recovery channel

$$\mathcal{R}_{\sigma,\mathcal{N}}(\mu)(b,d) = \mu(b)\,\sigma(d \mid b), \qquad \sigma(d \mid b) := \frac{\sigma_{BD}(b,d)}{\sigma_B(b)}.$$

Then $\mathcal{R}_{\sigma,\mathcal{N}}$ is stochastic, completely positive and trace preserving on the diagonal algebra, and ℓ^1-contractive:

$$\|\mathcal{R}_{\sigma,\mathcal{N}}(\mu) - \mathcal{R}_{\sigma,\mathcal{N}}(\nu)\|_1 \le \|\mu - \nu\|_1.$$

The support inverse is uniformly bounded on this domain by

$$\|\sigma_B^{-1/2}\| \le \gamma_\sigma^{-1/2}.$$

If a collar state has the exact classical Markov form

$$\omega_{ABD}(a,b,d) = \omega_{AB}(a,b)\sigma(d \mid b),$$

then $(\mathrm{id}_A \otimes \mathcal{R}_{\sigma,\mathcal{N}})(\omega_{AB}) = \omega_{ABD}$. The support obstruction is exact: if $\sigma_B(b) = 0$ and an input assigns mass to b, the Petz inverse on that sector is undefined unless the channel domain is restricted or a separate trace-preserving completion is declared.

Proof. The displayed formula is the finite diagonal specialization of the Petz map

$$\sigma_{BD}^{1/2}\left(\sigma_B^{-1/2}\mu\,\sigma_B^{-1/2} \otimes \mathbf{1}_D\right)\sigma_{BD}^{1/2}.$$

Full support of σ_B makes the inverse well defined, with support gap $\gamma_\sigma > 0$, hence $\|\sigma_B^{-1/2}\| \le \gamma_\sigma^{-1/2}$. Nonnegativity is immediate, and

$$\sum_{b,d} \mathcal{R}_{\sigma,\mathcal{N}}(\mu)(b,d) = \sum_b \mu(b) \sum_d \sigma(d \mid b) = \sum_b \mu(b),$$

so the map is trace preserving. Diagonal positive trace-preserving maps are completely positive on the diagonal algebra. For signed diagonal inputs,

$$\|\mathcal{R}_{\sigma,\mathcal{N}}(\mu) - \mathcal{R}_{\sigma,\mathcal{N}}(\nu)\|_1 = \sum_{b,d} |\mu(b) - \nu(b)|\, \sigma(d \mid b) = \sum_b |\mu(b) - \nu(b)|,$$

which gives the stated contraction. The exact Markov recovery identity follows by substitution. If $\sigma_B(b) = 0$, the factor $\sigma_B^{-1/2}$ has no value on that sector; mass placed there by an input is outside the Petz support domain. That is the claimed obstruction. □

Proposition 3.17 (Accepted repair moves are decreasing)**.** *For the accepted transactional repair law of Definitions 3.1–3.4, every enabled repair strictly decreases the declared exact measure:*

$$s \to t \implies \mu(t) \prec \mu(s).$$

On the scalar Φ-branch, this specializes to $\Phi(t) < \Phi(s)$.

Proof. This is part of the commit validation in Definition 3.4. For a single-site scalar-Φ transaction it reduces to the touched-overlap contract of Definition 3.2; for an aggregate conflict component it is checked on the aggregate payload before the component can commit. □

Proposition 3.18 (Termination from the OPH Lyapunov functional)**.** *Under Proposition 3.17, every repair sequence is finite; equivalently, the repair relation $\to$ is terminating.*

Proof. Along any nontrivial repair step $s \to t$, Proposition 3.17 gives $\mu(t) \prec \mu(s)$. The measure order is well founded, so an infinite strictly descending chain is impossible. On the scalar-Φ branch this is the older finite-value argument on $\Phi(\Sigma)$. □

Proposition 3.19 (Finite repair step bound)**.** *Let $s_0 \in \Sigma$. Under Proposition 3.17, every accepted repair run starting at s_0 has length at most the number of distinct μ-values reachable below $\mu(s_0)$ minus one. On the scalar-Φ branch this gives*

$$\big|\{\, \Phi(s) : s \in \Sigma,\ \Phi(s) \le \Phi(s_0) \,\}\big| - 1 \le |\Phi(\Sigma)| - 1.$$

If, additionally, Φ is integer-valued, or more generally every accepted move lowers Φ by at least a fixed $\eta > 0$, then the run length satisfies

$$T(s_0) \le \left\lceil \frac{\Phi(s_0)}{\eta} \right\rceil.$$

This is a finite descent bound in repair steps only. It is not a wall-clock bound, not a probability-one scheduling statement, and not spectral or exponential convergence.

Proof. The values of μ strictly decrease along the run, so no μ-value can occur twice. This gives the finite reachable-value bound. On the scalar-Φ branch, the values of Φ strictly decrease along the run, so no value in $\{\, \Phi(s) : s \in \Sigma,\ \Phi(s) \le \Phi(s_0) \,\}$ can occur twice. This gives the finite value-set bound. If each move lowers Φ by at least η, after T moves the value has dropped by at least $T\eta$. Since $\Phi \ge 0$, $T\eta \le \Phi(s_0)$, giving the displayed ceiling bound. □

Remark 3.20 (Termination is not confluence). Proposition 3.18 proves only that accepted repair runs stop. It does not prove that two update orders stop at the same physical state. Schedule-independence enters only after the local-diamond property of Proposition 3.21 is combined with termination via Newman's lemma and with repair completeness in Assumption 3.12. If two accepted repair schedules from the same initial state reach different observer-facing quotient normal forms, with no declared holonomy or higher-gauge obstruction and no mere gauge-representative difference, then the proposed repair law is not OPH-admissible as a consensus mechanism.

Proposition 3.21 (Local confluence from atomic conflict-component commits)**.** *For the accepted repair law of Definitions 3.1–3.4, the transactional repair relation is locally confluent.*

Proof. Take a one-step peak $t \leftarrow s \rightarrow u$, and let the two committed aggregate transactions come from enabled conflict components K and L at s. If $K = L$, the canonical component rule in Definition 3.4 gives the same aggregate payload, so $t = u$.

Assume $K \neq L$. Then no write set in one component intersects any read-or-write set in the other. Therefore each transaction's snapshot remains valid after the other commits, and the unchanged-outside-write test is unaffected. Validation-complete read sets ensure that every measure term whose value can change under one component was read by that component, so the descent check made at s remains the same after the other disjoint component is committed. Boundary and sector preservation are likewise checked on disjoint protected supports or on unchanged data. Thus each transaction can be committed after the other, and the two applications write disjoint registers to the same values. The two length-two paths therefore meet at an identical state v. This is the local diamond. $\square$

Theorem 3.22 (Asynchronous confluence / fixed-point law)**.** *For the accepted repair law of Definitions 3.1–3.4, under Assumption 3.12, each fixed initial state $s \in \Sigma$ has a unique normal form*

$$\mathrm{nf}_\lambda(s) \in C,$$

and every maximal asynchronous repair execution from s terminates at that same state. The terminal state is independent of update order because descent, local confluence on the physical quotient, and repair completeness all hold; termination alone is not enough.

Proof. By Proposition 3.18, the repair relation $\rightarrow$ is terminating. By Proposition 3.21, it is locally confluent. Newman's lemma [3] therefore makes it confluent. A terminating confluent repair relation has a unique normal form reachable from each fixed initial state. By Assumption 3.12, the normal forms are exactly C. Every maximal execution terminates and reaches $\mathrm{nf}_\lambda(s)$, and that terminal state is independent of update order. $\square$

Corollary 3.23 (Objective law is schedule-independent)**.** *Let $M : \Sigma \rightarrow Y$ be any observable. Under the hypotheses of Theorem 3.22, $M(\mathrm{nf}_\lambda(s))$ is independent of the asynchronous update schedule. If physical law is identified with the map $s \mapsto M(\mathrm{nf}_\lambda(s))$, then physical law is objective.*

Proof. All schedules from the same initial s terminate at $\mathrm{nf}_\lambda(s)$, so all yield the same M-value. $\square$

Here objectivity is identified with schedule-independent convergence of the repair dynamics; Theorem 5.17 sharpens this to schedule-independent convergence of the physical observable algebra even when microscopic representatives are not unique.

Remark 3.24 (Quantifier on normal-form uniqueness)*.* Theorem 3.22 proves uniqueness from a fixed initial state, and Theorem 5.2 below turns that into uniqueness from a fixed physical quotient state. It does not say that all possible initial data settle to one universal state. Different boundary conditions, conserved charges, root packets, holonomy sectors, or external records can legitimately determine different normal forms.

In the terminology of the repository file `CONSISTENCY_STACK.md`, the quotient normal-form theorems of this section—Theorem 3.9, Theorem 3.22, and the boundary-conditioned uniqueness statement of Theorem 5.4 below—are the instantiation of stack row C2 (overlap consensus): the public world is the quotient normal form that survives agreement on overlaps. The row is falsifiable at this theorem level, and two failure shapes would break it: an OPH-admissible overlap presentation on which no quotient normal form exists under the declared descent, local-diamond, and repair-completeness conditions, or two inequivalent quotient normal forms surviving from the same declared boundary data where the fiber-uniqueness hypothesis of Theorem 5.4 holds. Either exhibit defeats C2 on its declared branch; neither has been produced.

4 Why Local Agreement Is Not Enough: Cycle Holonomy and Frustration

Pairwise neighbor agreement does not by itself imply global consistency.

Theorem 4.1 (Cycle-obstruction / holonomy criterion)**.** *Let A be an abelian group, and let $G = (V, E)$ be a connected graph with an arbitrary orientation on each edge. For each oriented edge $e : u \to v$, assign a label $b_e \in A$. Consider the affine consistency equations*

$$x_v - x_u = b_e \qquad \textit{for every oriented edge } e : u \to v,$$

where $x_v \in A$ are unknown patch labels. A global solution $x : V \to A$ exists if and only if for every cycle $C \subseteq G$,

$$\sum_{e \in C} \varepsilon_C(e)\, b_e = 0,$$

where $\varepsilon_C(e) = +1$ if the cycle traverses e in the chosen orientation and -1 otherwise.

Proof. **Necessity.** Suppose x is a solution. Summing the edge equations around any cycle C,

$$\sum_{e:u\to v \in C} \varepsilon_C(e)\,(x_v - x_u) = \sum_{e \in C} \varepsilon_C(e)\, b_e.$$

The left side telescopes to 0 because every vertex appears once with $+$ sign and once with $-$ sign.

Sufficiency. Fix a root $r \in V$. For any vertex v, choose a path $P_{r\to v}$ and define

$$x_v := \sum_{e \in P_{r\to v}} \varepsilon_{P_{r\to v}}(e)\, b_e, \qquad x_r := 0.$$

If P and P' are two paths from r to v, traversing P followed by the reverse of P' yields a cycle. By the vanishing-holonomy assumption, the total signed sum is zero, so x_v is well-defined. For any edge $e : u \to v$, extending a path to u by that edge gives $x_v = x_u + b_e$. □

Corollary 4.2 (Parity triangle: pairwise consistency is not enough)**.** *Take $A = \mathbb{Z}_2$ on the triangle A-B-C-A with edge labels $b_{AB} = 0$, $b_{BC} = 0$, $b_{CA} = 1$. Each individual edge equation is satisfiable. But the global system is not: the cycle sum is $0 \oplus 0 \oplus 1 = 1 \neq 0$.*

This example shows that pairwise consistency does not imply a global solution. The obstruction is carried by the cycle.

Corollary 4.3 (Stable defects as frustrated holonomy)**.** *Define the defect energy*

$$\Phi_b(x) := \sum_{e:u\to v \in E} w_e\, \mathbf{1}[x_v - x_u \neq b_e]\,.$$

If the cycle-holonomy condition fails, then $\min_{x:V\to A} \Phi_b(x) > 0$. Every minimizer contains irreducible residual inconsistency.

Proof. If $\min_x \Phi_b(x) = 0$, some assignment satisfies all edge equations, contradicting Theorem 4.1. $\square$

Residual inconsistencies of this type cannot be removed by local repair moves. In the OPH interpretation they are stable topological defects of the reconciliation dynamics.

Theorem 4.4 (Higher-gauge defect hierarchy). *Let a finite overlap nerve carry crossed-module defect data (g_{ij}, h_{ijk}) for a compact crossed module*

$$H \xrightarrow{\partial} G.$$

Under local rechartings by

$$C^1(N,H) \rtimes C^0(N,G),$$

the nonabelian Čech class

$$q = [(g,h)] \in \check{H}^2(N, H \to G)$$

is invariant and classifies the full crossed-module orbit. Its neutral element is equivalent to full gauge trivialization of both levels. The higher associator alone is removable precisely when q lies in the image of the natural strict-locus map

$$\check{H}^1(N,G) \longrightarrow \check{H}^2(N, H \to G), \qquad [g] \longmapsto [(g,1)].$$

After such a strictification, endpoint-only ordinary reconciliation additionally requires an allowed strict representative with trivial represented loop holonomy. The strict-locus map need not be injective, so q need not determine one ordinary H^1 class.

Proof. The allowed rechartings are exactly the crossed-module coboundaries, so they preserve the full orbit. The neutral orbit admits the completely trivial representative. More generally, the higher associator is removed when the orbit meets the strict locus $(g,1)$; different points of that locus can be related by H-valued edge changes. Ordinary loop coherence is therefore the separate existential holonomy test stated above. $\square$

5 Gauge Symmetry as Implementation Hiding

Definition 5.1 (Gauge action). *Let $\Gamma = \prod_{i \in V} \Gamma_i$ act on Σ componentwise. The action is a **gauge action** if for every $e = \{i,j\} \in E$,*

$$\pi_{i,e}(\gamma_i \cdot x) = \pi_{i,e}(x) \qquad \forall\, x \in S_i,\ \forall\, \gamma_i \in \Gamma_i.$$

Gauge changes alter hidden local representations but do not alter overlap data.

Gauge transformations change hidden local representations while leaving overlap data fixed. Write

$$q : \Sigma \to \Sigma/\Gamma, \qquad q(s) = [s],$$

for the gauge-orbit map. A **physical repair law** is the family of quotient-local maps

$$\overline{T}_i^{\lambda} : \Sigma/\Gamma \to \Sigma/\Gamma$$

induced by the recovery-derived collar updates of Definition 3.1. A **representative repair family** is any choice of lifts

$$T_i^{\lambda} : \Sigma \to \Sigma$$

such that

$$q \circ T_i^\lambda = \overline{T}_i^\lambda \circ q.$$

This is the finite patch-net form of saying that the repair step is defined first on overlap-invariant physical data and only then lifted to hidden representatives. The rooted-tree packet domain of Theorem 3.14 proves repair completeness and quotient descent on a nontrivial exported packet net, while Proposition 3.16 proves the classical full-support Petz clause used by that domain. A broader fixed-cutoff branch requires repair completeness for the chosen exported packet net and the support/CPTP clause on the Petz branch where that channel is used. The touched-overlap local-fit contract gives scalar Φ-descent for primitive proposals on the corresponding branch, while the transaction layer supplies exact accepted-step descent. Proposition B.8 together with Definition 3.3 supplies the quotient-local compatibility package used to build canonical aggregate payloads; the local diamond then comes from atomic conflict-component commits. Stability of that package under refinement or branch change is a separate question. The point here is also that no extra gauge-covariance axiom is needed once repair is formulated on the quotient.

Theorem 5.2 (Gauge quotient theorem)**.** *Under the gauge action of Definition 5.1, any representative lift of a physical repair law as just defined, and the hypotheses of Theorem 3.22,*

$$q\big(\mathrm{nf}_\lambda(\gamma \cdot s)\big) = q\big(\mathrm{nf}_\lambda(s)\big) \qquad \forall\, \gamma \in \Gamma,\ \forall\, s \in \Sigma.$$

Hence the normal-form map descends to the quotient:

$$\overline{\mathrm{nf}}_\lambda : \Sigma/\Gamma \to \Sigma/\Gamma, \qquad [s] \mapsto [\mathrm{nf}_\lambda(s)].$$

Proof. Let $s \to t$ be an accepted aggregate transaction, and let $\overline{\tau}$ be its quotient payload. If $s' = \gamma \cdot s$, the representative-lift hypothesis gives an accepted lift $s' \to t'$ of the same quotient transaction, and

$$q(t') = \overline{\tau}\big(q(s')\big) = \overline{\tau}\big(q(s)\big) = q(t).$$

Thus gauge-equivalent inputs induce the same repaired orbit, and by induction the orbit reached after any repair sequence depends only on the initial orbit. Every maximal repair sequence from s ends at $\mathrm{nf}_\lambda(s)$ by Theorem 3.22, so the terminal orbit $q(\mathrm{nf}_\lambda(s))$ depends only on $q(s) = [s]$. This makes

$$[s] \longmapsto [\mathrm{nf}_\lambda(s)]$$

well-defined on Σ/Γ. □

Corollary 5.3 (Repair respects gauge)**.** *Let $Q = \Sigma/\Gamma$, let $q : \Sigma \to Q$ be the quotient map, and let*

$$\mathrm{Rep}_\lambda^\Sigma := \mathrm{Rep}_\lambda \circ q : \Sigma \to Q$$

be the quotient-valued physical repair of a representative. Then for every $\gamma \in \Gamma$ and $s \in \Sigma$,

$$\mathrm{Rep}_\lambda^\Sigma(\gamma \cdot s) = \mathrm{Rep}_\lambda^\Sigma(s).$$

Equivalently,

$$\mathrm{Rep}_\lambda(q(\gamma \cdot s)) = \mathrm{Rep}_\lambda(q(s)).$$

If $M : Q \to Y$ is any physical observable, then

$$M(\mathrm{Rep}_\lambda^\Sigma(\gamma \cdot s)) = M(\mathrm{Rep}_\lambda^\Sigma(s)).$$

Proof. Since q is the quotient map, $q(\gamma \cdot s) = q(s)$. Therefore

$$\mathrm{Rep}_\lambda^\Sigma(\gamma \cdot s) = \mathrm{Rep}_\lambda(q(\gamma \cdot s)) = \mathrm{Rep}_\lambda(q(s)) = \mathrm{Rep}_\lambda^\Sigma(s).$$

Applying M gives the observable statement. □

Theorem 5.4 (Boundary-conditioned quotient uniqueness)**.** *Let*

$$B : \Sigma/\Gamma \to \mathcal{B}$$

record fixed external boundary data, conserved charge, root packet, holonomy sector, or task input. Assume accepted quotient repairs preserve B:

$$[s] \to [t] \implies B([s]) = B([t]).$$

For $b \in \mathcal{B}$, define the consistent quotient fiber

$$C_b := \{\, x \in q(C) : B(x) = b \,\}.$$

If every C_b has at most one element, then all initial states with the same boundary value settle to the same observer-facing quotient normal form:

$$B([s]) = B([s']) \implies \overline{\mathrm{nf}}_\lambda([s]) = \overline{\mathrm{nf}}_\lambda([s']).$$

Proof. Theorem 5.2 gives unique quotient normal forms $\overline{\mathrm{nf}}_\lambda([s])$ and $\overline{\mathrm{nf}}_\lambda([s'])$ in $q(C)$. Boundary preservation along accepted repair sequences gives

$$B(\overline{\mathrm{nf}}_\lambda([s])) = B([s]), \qquad B(\overline{\mathrm{nf}}_\lambda([s'])) = B([s']).$$

If $B([s]) = B([s']) = b$, both quotient normal forms lie in C_b. By the unique consistent extension assumption, C_b has at most one element, so the quotient normal forms are equal. □

Remark 5.5 (Same-source and cross-source quantifiers)*.* Theorem 3.22 compares repair schedules from one fixed initial source. The stronger comparison of endpoints reached from different sources with the same protected boundary is controlled by the consistent boundary fibers. Under boundary preservation and NF $= q(C)$, Ref. [1] proves that cross-source endpoint agreement, modulo any declared silent equivalence, is equivalent to injectivity of the induced boundary map on the consistent quotient. Weak normalization and all-schedule liveness are separate obligations. The layered and functional carriers below establish the injectivity premise only on their named finite domains; same-source confluence alone does not establish it for an arbitrary physical boundary map.

5.1 A multi-edge finite carrier for boundary reconstruction

Definition 5.6 (Layered functional boundary carrier)**.** *A **layered functional boundary carrier** is a finite directed layered graph*

$$G = (V, E), \qquad V = L_0 \sqcup L_1 \sqcup \cdots \sqcup L_D,$$

with $D \geq 2$. The boundary layer is L_0. For each $v \in L_d$, $d \geq 1$, choose a nonempty parent set

$$P(v) \subseteq L_0 \sqcup \cdots \sqcup L_{d-1}$$

and a finite alphabet A_v. Each interior vertex has a deterministic local rule

$$F_v : \prod_{u \in P(v)} A_u \to A_v.$$

The carrier is genuinely multi-edge when at least two dependency edges occur and at least one layer contains more than one dependency or more than one repaired vertex.

The quotient state space is

$$Q = \prod_{v \in V} A_v,$$

and the boundary map is

$$B : Q \to \prod_{v \in L_0} A_v, \qquad B(a) = a|_{L_0}.$$

Given boundary data $b \in \prod_{v \in L_0} A_v$, define its functional extension $E(b) \in Q$ recursively by

$$E(b)_v = b_v \quad (v \in L_0),$$

and, for $v \in L_d$, $d \geq 1$,

$$E(b)_v = F_v\big((E(b)_u)_{u \in P(v)}\big).$$

Optional cross-check edges may be included by choosing finite predicates

$$\chi_e(a_u, a_v) = 0.$$

A state $a \in Q$ is consistent when every interior functional equation

$$a_v = F_v\big((a_u)_{u \in P(v)}\big)$$

holds and all cross-check predicates pass. Let C_Q be the set of consistent states. A boundary b is admissible when $E(b) \in C_Q$; equivalently,

$$C_Q = \{\, E(b) : b \text{ admissible} \,\}.$$

For each layer $d = 1, \ldots, D$, define a layer repair map $R_d : Q \to Q$ by

$$(R_d(a))_v = \begin{cases} F_v\big((a_u)_{u \in P(v)}\big), & v \in L_d, \\ a_v, & v \notin L_d. \end{cases}$$

The full staged repair sweep is

$$R_{\text{sweep}} = R_D \circ R_{D-1} \circ \cdots \circ R_1.$$

Gauge representatives may be added by replacing A_v with $A_v \times H_v$, letting Γ_v act only on H_v, and defining all boundary and consistency maps through the A_v-coordinate. The quotient is then the Q above.

Theorem 5.7 (Layered carrier proves $H_B \wedge H_{\text{fib}}$)**.** *Let G be a layered functional boundary carrier. For every initial quotient state $a \in Q$, set*

$$a^{(0)} = a, \qquad a^{(d)} = R_d(a^{(d-1)}), \quad d = 1, \ldots, D.$$

Let $b = B(a)$. Then

$$B(a^{(d)}) = b \qquad \text{for all } d = 0, \ldots, D.$$

If b is admissible, then

$$a^{(D)} = E(b) \in C_Q.$$

The consistent boundary fiber is a singleton:

$$C_Q \cap B^{-1}(b) = \{E(b)\}.$$

Thus preserved boundary data plus consistency reconstruct the observer-facing quotient bulk on this carrier.

Proof. Boundary preservation is immediate because each R_d writes only layer L_d, with $d \geq 1$. No R_d writes L_0, so $B(a^{(d)}) = B(a) = b$ for every stage.

We prove by induction on d that after stage d,

$$a_v^{(d)} = E(b)_v \qquad \text{for every } v \in L_0 \sqcup \cdots \sqcup L_d.$$

For $d = 0$, this is the definition of b. Assume the claim holds through layer $d-1$, and let $v \in L_d$. Every parent $u \in P(v)$ lies in an earlier layer, so $a_u^{(d-1)} = E(b)_u$. The layer repair therefore gives

$$a_v^{(d)} = F_v\big((a_u^{(d-1)})_{u \in P(v)}\big) = F_v\big((E(b)_u)_{u \in P(v)}\big) = E(b)_v.$$

Earlier layers are not modified by R_d, so the induction closes. At $d = D$, all layers agree with $E(b)$. If b is admissible, $E(b) \in C_Q$.

For boundary-fiber uniqueness, let $c \in C_Q \cap B^{-1}(b)$. Since c has boundary b, it agrees with $E(b)$ on L_0. Since $c \in C_Q$, every interior vertex satisfies

$$c_v = F_v((c_u)_{u \in P(v)}).$$

The same induction on layers gives $c_v = E(b)_v$ for every vertex. Hence $c = E(b)$. □

Corollary 5.8 (Boundary reconstruction by global Repair)**.** *Assume the layered carrier's layer updates are included among the accepted aggregate transactions of an OPH-admissible finite quotient repair presentation, and let $b = B(x)$ be admissible. Then*

$$\mathrm{Rep}_\lambda(x) = E(b).$$

Consequently, for every physical readout

$$\mathrm{Read} : Q \to \mathcal{Y}$$

that factors through the quotient normal form,

$$\mathrm{Read}(\mathrm{Rep}_\lambda(x)) = \mathrm{Read}(E(B(x))).$$

Proof. By Theorem 3.9,

$$\mathrm{Rep}_\lambda(x) \in C_Q \qquad \text{and} \qquad B(\mathrm{Rep}_\lambda(x)) = B(x) = b.$$

Therefore $\mathrm{Rep}_\lambda(x) \in C_Q \cap B^{-1}(b)$. Theorem 5.7 identifies this fiber with $\{E(b)\}$, so $\mathrm{Rep}_\lambda(x) = E(b)$. Applying Read gives the readout statement. □

5.2 Finite binary audit fixture: a sharp reconstruction carrier

The layered carrier above proves boundary reconstruction for a feed-forward finite class. A useful stress test for the same logic is a two-neighbor linear binary update on a finite cylinder. A row is a function

$$x_t : \mathbb{Z}_n \to \mathbb{F}_2,$$

and the update rule is

$$x_{t+1}(j) = x_t(j-1) + x_t(j+1) \pmod 2.$$

This update is not a proposed microscopic physics law. It is a minimal finite-consensus fixture: the local rule is linear, all records are exact finite objects, and boundary reconstruction can be tested without continuum or geometric assumptions.

For the time slab $0, \dots, t$, a two-column timelike tube

$$\{j_0, j_0 + g\} \times \{0, \dots, t\}$$

is an information set when its values determine the whole finite spacetime record. Matscheko's OPHProofChain Lean audit proves the sharp two-column classification:

$$\text{tube}(g) \text{ is an information set} \quad \Longleftrightarrow \quad \gcd(g, n) = 1 \text{ and } n \le 2(t+1).$$

Thus adjacent columns ($g = 1$) achieve the full capacity threshold, while a non-coprime stride loses information at every horizon. The same fixture also separates realizable from unrealizable boundary readings: if a tube reading is not carried by any consistent record, no boundary-preserving repair operator can make it consistent without changing the boundary.

The audit then assembles a Route-A repair on this same carrier. The positive operator is a local transactional decoder: each transaction writes one non-tube cell using a bounded edge-local window, the declared rank schedule terminates in one finite pass, and the tube reading is preserved at every accepted step. At the sharp threshold $n \le 2(t+1)$, any two records with the same tube reading settle to the same normal form; the settled world is consistent exactly when the tube fiber is realizable.

The same formal fixture supplies the important negative controls. On this binary cylinder there is no single-patch frustration-free local repair satisfying the strongest $H_1 \wedge H_2 \wedge H_3$ binder form. The canonical single-patch repair can also stall on the smallest audit witness $(n, t) = (3, 2)$, leaving a broken edge in normal form. This is not a failure of the positive decoder; the stalled record has an unrealizable tube fiber. The lesson for OPH is structural: boundary reconstruction is theorem-grade only after the preserved boundary, realizable-fiber condition, and declared repair roster are specified. A bare local-update slogan is too weak.

Theorem 5.9 (Functional selected-fiber uniqueness). *Let $Q_b := B^{-1}(b) \subseteq \Sigma/\Gamma$ be a same-boundary quotient fiber presented on a rooted finite overlap graph with spanning tree T, root r, and patch state sets X_v. Assume the boundary value fixes the root value*

$$u_r = \beta(b),$$

and every non-root vertex has a deterministic extension map

$$f_v : X_{p(v)} \times \mathcal{B} \to X_v.$$

Define u_b recursively by

$$u_v = f_v(u_{p(v)}, b).$$

Evaluate all non-tree overlap equations, sector constraints, and holonomy checks on this candidate. If they all pass, then $C_b = \{u_b\}$. If any check fails, then $C_b = \varnothing$.

Proof. Every consistent state in Q_b must have root value $\beta(b)$. Along each tree edge, the deterministic extension equation forces the child value to be $f_v(u_{p(v)}, b)$. Induction over tree depth therefore forces every consistent state in the fiber to equal the single recursively constructed candidate u_b. The remaining constraints are exactly the non-tree overlaps, sector equations, and holonomy checks. If they pass, u_b is a consistent state and no other state can be. If one fails, the only tree-compatible candidate is inconsistent, so no element of C_b exists. □

The layered functional carrier of Theorem 5.7 is the finite multi-edge, multi-step witness that the boundary-fiber hypothesis is non-vacuous beyond one-dimensional propagation intuition.

Proposition 5.10 (Tree repair realizes the selected extension). *Under the hypotheses of Theorem 5.9, choose positive weights*

$$w_v > \sum_{c:p(c)=v} w_c$$

and set

$$\Phi_T(x) = \sum_{v \neq r} w_v \, \mathbf{1}[x_v \neq f_v(x_{p(v)}, b)].$$

The repair that sets a non-root vertex to $f_v(x_{p(v)}, b)$ strictly decreases Φ_T, and the aggregate transaction for a connected tree-repair conflict component computes the same result in increasing tree depth.

Proof. Repairing v removes the v-term of weight w_v. Only children of v can become unmatched, contributing at most $\sum_{c:p(c)=v} w_c$, which is strictly smaller than w_v. Thus the tree potential decreases. For a connected conflict component, applying parent repairs before child repairs is forced by the same extension equations, and any different parenthesization has the same quotient result because each child value is a deterministic function of the repaired parent and b. □

Corollary 5.11 (Nontrivial branch elimination). *Assume transactional confluence, boundary preservation, repair completeness, and Theorem 5.9. If a selected boundary fiber has $|Q_b| \geq 2$ and $C_b = \{u_b\}$, then*

$$\overline{\mathrm{nf}}_\lambda(Q_b) = \{u_b\},$$

while $Q_b \setminus C_b \neq \varnothing$. Thus multiple candidate interiors exist, inconsistent candidates are eliminated by repair, and all surviving candidates share one quotient normal form. If $C_b = \varnothing$, the branch is obstructed. If a union solver returns two physically distinct consistent quotient endpoints for the same selected data, the correct label is AMBIGUOUS-UNION-REPAIR *or an explicit continuation gate, not selection by hash.*

Remark 5.12 (Receipts do not replace the theorem). Implementation receipts named seam descent, atomic commit, distributed local diamond, repair completeness, same-boundary multistart confluence, and quotient normal-form canonical hash are public evidence contracts for the finite theorem hypotheses above. They are not proof substitutes. In particular, a normal-form hash certifies that two emitted quotient normal forms agree after the declared quotienting; it is not allowed to choose between two physically distinct minimizing states. The same convention applies to continuation receipts used by implementation evidence. A receipt is a compact name for primitive evidence and residual checks. Producer-declared booleans, labels, hashes, or successful plots cannot override failed residuals, missing objects, or undeclared quotient maps. When a later paper names a physical receipt, the evidence bundle must expose the raw objects needed to recompute it.

Remark 5.13 (Triality as finite quotient-fusion example). The $E_8/\mathrm{Spin}(8)$ triality certificate is a finite exact example of the quotient discipline used here. Before quotienting by the allowed outer symmetry, the vector $\mathrm{Alt}(9)$ and positive-half-spin $2{\cdot}\mathrm{Alt}(9)$ presentations have different orbit partitions on $E_8/2E_8 \setminus \{0\}$, namely $\{9, 36, 84, 126\}$ and $\{120, 135\}$, so they are not conjugate in $O_8^+(2)$. After adjoining Spin(8) triality they become different charts of one triality-fused exceptional datum. This is not a new confluence theorem and does not imply repair termination, noisy convergence, code distance, BFT liveness, or hardware speedup.

Corollary 5.14 (Gauge-invariant law). *If $M : \Sigma \to Y$ is gauge-invariant ($M(\gamma \cdot s) = M(s)$ for all γ), then $M(\mathrm{nf}_\lambda(s))$ depends only on the gauge orbit $[s]$, not the representative.*

Proof. Because M is gauge-invariant, it factors through the orbit map: $M = \overline{M} \circ q$ for some $\overline{M} : \Sigma/\Gamma \to Y$. Theorem 5.2 gives

$$q\big(\mathrm{nf}_\lambda(\gamma \cdot s)\big) = q\big(\mathrm{nf}_\lambda(s)\big),$$

hence

$$M\big(\mathrm{nf}_\lambda(\gamma \cdot s)\big) = \overline{M}\big(q(\mathrm{nf}_\lambda(\gamma \cdot s))\big) = \overline{M}\big(q(\mathrm{nf}_\lambda(s))\big) = M\big(\mathrm{nf}_\lambda(s)\big).$$

□

Remark 5.15 (Sphere folding as quotient readout). When a spherical support chart $\chi_{S,r}$ is supplied by the screen-microphysics branch, the folded screen presentation of a finite state s is

$$\mathrm{Fold}_{S,r}(s) = \chi_{S,r}(n_r(\pi_r(s)))\,.$$

Here π_r passes to the physical quotient and n_r is the accepted repair normal-form map. The word "folding" therefore names a readout of the same quotient normal form studied in this paper. It does not add a repair law, force term, or hidden state variable.

Definition 5.16 (Physical observable algebra on the quantum lift). *Fix a finite patch region or union collar R on the declared fixed-cutoff quantum lift of Appendix B. Its* ***physical observable algebra*** *$\mathcal{A}_{\mathrm{phys}}(R)$ is the fixed-point collar algebra under the compact boundary redundancy action on the ordinary or central-defect branch, or the corresponding quotient-local algebra on the genuinely noncentral branch. The central sector projectors carried by the collar decomposition belong to $Z(\mathcal{A}_{\mathrm{phys}}(R))$. A* ***physical observable*** *is any $X \in \mathcal{A}_{\mathrm{phys}}(R)$. For a microscopic representative $s \in \Sigma$, let ω_R^s denote the induced state on $\mathcal{A}_{\mathrm{phys}}(R)$.*

Theorem 5.17 (Observable-level confluence on the quantum lift). *Under the hypotheses of Theorems 3.22 and 5.2, every initial orbit $[s] \in \Sigma/\Gamma$ has a unique quotient normal form*

$$\overline{\mathrm{nf}}_\lambda([s]) \in q(C).$$

Fix a declared region R. Let $t, u \in \Sigma$ be terminal microscopic representatives reached by representative lifts of maximal repair sequences from initial states in the same orbit $[s]$. If the induced R-collar states of t and u are representatives of the same quotient-local glued state in the sense of Proposition B.8, then for every physical observable $X \in \mathcal{A}_{\mathrm{phys}}(R)$,

$$\omega_R^t(X) = \omega_R^u(X).$$

Hence all physical observables converge to the same overlap-consistent values even when the microscopic terminal representatives differ by gauge relabelings globally or by sector/higher-gauge relabelings inside one declared quotient-local glued state.

Proof. Theorems 3.22 and 5.2 give the unique quotient normal form $\overline{\mathrm{nf}}_\lambda([s]) = [\mathrm{nf}_\lambda(s)] \in q(C)$. Let t and u be as stated. By Corollary B.10, representatives of the same quotient-local glued state induce the same state on the physical observable algebra $\mathcal{A}_{\mathrm{phys}}(R)$, including the central sector projectors carried by that algebra. Therefore $\omega_R^t(X) = \omega_R^u(X)$ for every $X \in \mathcal{A}_{\mathrm{phys}}(R)$. □

Remark 5.18 (Inputs and boundary for observable-level confluence). Theorem 5.17 does not add a new repair hypothesis beyond the confluence and fixed-cutoff gluing package. Its inputs are exactly the representative-level confluence theorem, quotient descent of the repair law, Definition 5.16, and Corollary B.10 from the fixed-cutoff union-collar gluing package. The boundary item is extension of the same statement to broader refinement-stable branches where the declared union-collar compatibility is only approximate or where the chosen physical observable algebra itself changes under refinement.

Corollary 5.19 (Inert ancillary refinement does not change physical law)**.** *Let $K = \prod_i K_i$ be a finite ancillary state space and define $\Sigma^\eta := \Sigma \times K$. Lift the repair maps by*

$$T_i^{\lambda,\eta}(s,k) := (T_i^\lambda(s), k).$$

If $M^\eta : \Sigma^\eta \to Y$ ignores the ancillary factor,

$$M^\eta(s,k) = M(s),$$

with M gauge-invariant on Σ, then

$$M^\eta(\mathrm{nf}_\lambda^\eta(s,k)) = M(\mathrm{nf}_\lambda(s))$$

for all $(s,k) \in \Sigma^\eta$.

Proof. Because the ancillary factor is inert,

$$\mathrm{nf}_\lambda^\eta(s,k) = (\mathrm{nf}_\lambda(s), k).$$

Therefore $M^\eta(\mathrm{nf}_\lambda^\eta(s,k)) = M(\mathrm{nf}_\lambda(s))$, and Corollary 5.14 supplies gauge-orbit independence. □

Physical uniqueness therefore holds on the quotient by gauge or implementation hiding. The same statement is unchanged under inert ancillary stabilization.

Definition 5.20 (Finite packet closure simplex)**.** *Assume the finite fixed-cutoff consensus branch of Theorems 3.22 and 5.2, so the physical quotient $Q := \Sigma/\Gamma$ is finite and carries the schedule-independent quotient normal-form map*

$$\overline{\mathrm{nf}}_\lambda : Q \to Q.$$

Let

$$N_\lambda := \overline{\mathrm{nf}}_\lambda(Q) = q(C)$$

*be the quotient normal-form set. The **finite packet simplex** on Q is*

$$\Delta(Q) := \left\{ \mu : Q \to \mathbb{R}_{\geq 0} \;\middle|\; \sum_{x \in Q} \mu(x) = 1 \right\},$$

*and the **finite packet closure map** is the pushforward*

$$\mathcal{C}_\lambda : \Delta(Q) \to \Delta(Q), \qquad \mathcal{C}_\lambda(\mu) := (\overline{\mathrm{nf}}_\lambda)_* \mu,$$

equivalently

$$\mathcal{C}_\lambda(\mu)(y) = \sum_{\substack{x \in Q \\ \overline{\mathrm{nf}}_\lambda(x) = y}} \mu(x).$$

Theorem 5.21 (Finite packet-quotient closure map)**.** *On the finite fixed-cutoff consensus branch of Definition 5.20, the map $\mathcal{C}_\lambda$ is an affine continuous idempotent self-map of $\Delta(Q)$. Its image is exactly the normal-form simplex*

$$\Delta(N_\lambda) = \{\mu \in \Delta(Q) \mid \operatorname{supp}(\mu) \subseteq N_\lambda\}.$$

Hence the fixed points of $\mathcal{C}_\lambda$ are exactly the packets supported on quotient normal forms. For every initial quotient state $[s] \in Q$,

$$\mathcal{C}_\lambda(\delta_{[s]}) = \delta_{\overline{\mathrm{nf}}_\lambda([s])}.$$

Proof. Because $\overline{\mathrm{nf}}_\lambda : Q \to Q$ is a set map on a finite set, its pushforward on probability packets is affine and continuous in the finite-dimensional simplex topology.

By Theorems 3.22 and 5.2, the quotient normal form is schedule-independent and terminal, so

$$\overline{\mathrm{nf}}_\lambda \circ \overline{\mathrm{nf}}_\lambda = \overline{\mathrm{nf}}_\lambda.$$

Pushforward therefore gives

$$\mathcal{C}_\lambda^2 = \left(\overline{\mathrm{nf}}_\lambda\right)_* \left(\overline{\mathrm{nf}}_\lambda\right)_* = \left(\overline{\mathrm{nf}}_\lambda \circ \overline{\mathrm{nf}}_\lambda\right)_* = \mathcal{C}_\lambda,$$

so $\mathcal{C}_\lambda$ is idempotent.

If $\nu = \mathcal{C}_\lambda(\mu)$, then every point in $\operatorname{supp}(\nu)$ lies in the image of $\overline{\mathrm{nf}}_\lambda$, namely N_λ. Thus $\operatorname{im}(\mathcal{C}_\lambda) \subseteq \Delta(N_\lambda)$. Conversely, if $\nu \in \Delta(N_\lambda)$, then $\overline{\mathrm{nf}}_\lambda(y) = y$ for every $y \in N_\lambda$, so

$$\mathcal{C}_\lambda(\nu) = \nu.$$

Hence $\Delta(N_\lambda) \subseteq \operatorname{im}(\mathcal{C}_\lambda)$, proving $\operatorname{im}(\mathcal{C}_\lambda) = \Delta(N_\lambda)$. The same identity shows that ν is a fixed point if and only if it is supported on N_λ. The Dirac-mass formula is the pushforward of a point mass under $\overline{\mathrm{nf}}_\lambda$. □

Remark 5.22 (Boundary of the finite closure theorem)*.* Theorem 5.21 is an exact finite-branch result. It proves a closure map only on the finite packet simplex built from one fixed quotient carrier whose repair law is terminating, confluent, and quotient-descended. It does not prove a habitat-level closure map for arbitrary OPH state-and-law data, does not identify a nonempty observer-supporting invariant sector inside the Appendix-B habitat, and does not supply uniqueness or stability estimates beyond this finite packet branch.

6 Refinement-Limit Consensus Classes

The finite consensus theorem is the operational object used by the continuum branches. To pass from one finite patch net to a refining family, the paper uses an explicit inverse-limit package with named compatibility clauses.

Definition 6.1 (Separated cofinal refinement consensus system)**.** *Let $(R, \preceq)$ be a directed refinement set. For each $r \in R$, let*

$$Q_r := \Sigma_r / \Gamma_r$$

be the finite physical quotient state space of a patch presentation, let

$$n_r : Q_r \to Q_r$$

be the finite-stage quotient normal-form map supplied by Theorems 3.22 and 5.2, and let

$$h_r : Q_r \to \mathcal{H}_r$$

be the finite-stage holonomy or higher-gauge obstruction map supplied by Theorems 4.1 and 4.4. For $r \preceq s$, assume restriction maps

$$\rho_{sr} : Q_s \to Q_r, \qquad \chi_{sr} : \mathcal{H}_s \to \mathcal{H}_r,$$

with $\rho_{rr} = \mathrm{id}$, $\chi_{rr} = \mathrm{id}$, and the cocycle identities

$$\rho_{tr} = \rho_{sr} \circ \rho_{ts}, \qquad \chi_{tr} = \chi_{sr} \circ \chi_{ts} \qquad (r \preceq s \preceq t).$$

The system is a separated cofinal refinement consensus system when:

1. ***Normal-form naturality:***

$$\rho_{sr} \circ n_s = n_r \circ \rho_{sr} \qquad (r \preceq s).$$

2. ***Holonomy naturality:***

$$\chi_{sr} \circ h_s = h_r \circ \rho_{sr} \qquad (r \preceq s).$$

3. ***Visible separation:*** *two compatible families in $\varprojlim Q_r$, or in $\varprojlim \mathcal{H}_r$, are equal whenever their projections agree on a cofinal subset of stages.*

Theorem 6.2 (Refinement-limit consensus and holonomy classes)**.** *For any separated cofinal refinement consensus system, the formulas*

$$n_\infty((x_r)_r) := (n_r(x_r))_r, \qquad h_\infty((x_r)_r) := (h_r(x_r))_r$$

define maps

$$n_\infty : \varprojlim Q_r \to \varprojlim Q_r, \qquad h_\infty : \varprojlim Q_r \to \varprojlim \mathcal{H}_r.$$

The class $n_\infty(x)$ is the unique schedule-independent refinement-limit normal form of x. The class $h_\infty(x)$ is the refinement-limit holonomy obstruction. The pair $(n_\infty(x), h_\infty(x))$ is the refinement-limit consensus class. In the inverse-limit topology, finite-stage normal forms and holonomy classes converge to these two classes: for every stage r, all refinements $s \succeq r$ restrict to the fixed values

$$\rho_{sr}(n_s(x_s)) = n_r(x_r), \qquad \chi_{sr}(h_s(x_s)) = h_r(x_r).$$

The relation $h_\infty(x) = 0$ holds if and only if every finite projection has zero finite-stage obstruction. If $h_\infty(x) \neq 0$, visible separation gives a finite stage that witnesses the nonzero obstruction. If two refinement-limit candidates have the same normal-form and holonomy projections on a cofinal tail, then they determine the same refinement-limit consensus class.

Proof. Let $x = (x_r)_r \in \varprojlim Q_r$, so $\rho_{sr}(x_s) = x_r$ for $r \preceq s$. Normal-form naturality gives

$$\rho_{sr}(n_s(x_s)) = n_r(\rho_{sr}(x_s)) = n_r(x_r),$$

so $(n_r(x_r))_r$ is a compatible family and n_∞ is well defined. The same argument with holonomy naturality gives

$$\chi_{sr}(h_s(x_s)) = h_r(\rho_{sr}(x_s)) = h_r(x_r),$$

so h_∞ is well defined.

Each finite $n_r(x_r)$ is independent of update schedule by Theorems 3.22 and 5.2. Therefore every finite projection of $n_\infty(x)$ is schedule-independent, and visible separation makes the inverse-limit class unique. The displayed restriction identities are exactly the cylinder convergence statement in the inverse-limit topology.

The zero-obstruction statement follows from the definition of the inverse-limit zero class. The identity $h_\infty(x) = 0$ holds exactly when every projection $h_r(x_r)$ is zero. If $h_\infty(x) \neq 0$, at least one finite projection is nonzero, and any cofinal tail containing a refinement of that stage carries the same nonzero projected obstruction by holonomy naturality. The last claim is visible separation applied to the compatible normal-form and holonomy families. □

Remark 6.3 (Scope of the refinement theorem). Theorem 6.2 is a controlled continuum bridge. It proves persistence, exhaustion, and collapse of the consensus data once the OPH refinement system supplies the restriction maps and the two naturality clauses in Definition 6.1. It leaves uniform complexity bounds, automatic fair-block noisy-consensus certificates, and automatic normal-form naturality for arbitrary changing repair laws as separate branch conditions.

Definition 6.4 (Repair morphism)**.** *Let $(Q_s, \to_s, C_s, n_s)$ and $(Q_r, \to_r, C_r, n_r)$ be two finite quotient repair systems covered by Theorems 3.22 and 5.2. A map*

$$\rho_{sr} : Q_s \to Q_r$$

is a ***repair morphism*** *when:*

(i) every fine repair step $x \to_s y$ maps to a coarse repair path or a stutter,

$$\rho_{sr}(x) \to_r^* \rho_{sr}(y);$$

(ii) fine consistency maps into coarse consistency,

$$\rho_{sr}(C_s) \subseteq C_r.$$

For a concrete local repair law this is checked by giving, for every fine repair generator, a coarse witness word in the accepted coarse repair generators or the empty word, and by verifying that the restriction maps carry every fine consistency equation to a coarse consistency equation.

Theorem 6.5 (Normal-form naturality from repair morphisms)**.** *If $\rho_{sr} : Q_s \to Q_r$ is a repair morphism, then*

$$\rho_{sr} \circ n_s = n_r \circ \rho_{sr}.$$

Proof. Fix $x \in Q_s$. Since $x \to_s^* n_s(x)$, repeated use of the step-simulation clause gives

$$\rho_{sr}(x) \to_r^* \rho_{sr}(n_s(x)).$$

The consistency-preservation clause gives $\rho_{sr}(n_s(x)) \in C_r$, because $n_s(x) \in C_s$. Thus $\rho_{sr}(n_s(x))$ is a coarse normal form reachable from $\rho_{sr}(x)$. The coarse system has a unique normal form reachable from each initial quotient state, so $\rho_{sr}(n_s(x)) = n_r(\rho_{sr}(x))$. □

Definition 6.6 (Holonomy cochain morphism)**.** *For obstruction maps $h_s : Q_s \to \mathcal{H}_s$ and $h_r : Q_r \to \mathcal{H}_r$, a map $\chi_{sr} : \mathcal{H}_s \to \mathcal{H}_r$ is a* ***holonomy cochain morphism*** *when the edge, cycle, or crossed-module cochains used to compute h_s restrict to the cochains used to compute h_r, modulo the declared quotient identifications. Equivalently,*

$$\chi_{sr} \circ h_s = h_r \circ \rho_{sr}.$$

Remark 6.7 (How exact refinement naturality is discharged). Definition 6.1 may be used as an interface contract, but on an implemented exact branch the normal-form part is proved by Theorem 6.5 from explicit coarse witness words for fine repairs. The holonomy part is proved by the cochain morphism check of Definition 6.6. Approximate RG branches use the controlled defects of Definition 6.8 instead.

Definition 6.8 (Controlled coarse-graining / reconciliation square). *Fix refinement stages $r \preceq s$. Let Q_r, Q_s be the physical quotient state spaces, n_r, n_s their finite-stage normal-form maps, and h_r, h_s their holonomy or higher-gauge obstruction maps. Let*

$$\rho_{sr} : Q_s \to Q_r, \qquad \chi_{sr} : \mathcal{H}_s \to \mathcal{H}_r$$

be the coarse-graining maps on quotient states and obstruction data. Equip Q_r and $\mathcal{H}_r$ with pseudometrics d_r^Q and $d_r^{\mathcal{H}}$ that define the macroscopic readout scale at stage r.

The square is $(\varepsilon_{sr}^n, \varepsilon_{sr}^h)$-controlled when, for every $x \in Q_s$,

$$d_r^Q(\rho_{sr}(n_s(x)),\, n_r(\rho_{sr}(x))) \le \varepsilon_{sr}^n,$$

and

$$d_r^{\mathcal{H}}(\chi_{sr}(h_s(x)),\, h_r(\rho_{sr}(x))) \le \varepsilon_{sr}^h.$$

The errors are ***cofinally vanishing*** *when, for every fixed macroscopic stage r and every $\delta > 0$, there is a refinement stage $s_0 \succeq r$ such that $\varepsilon_{sr}^n, \varepsilon_{sr}^h < \delta$ for all $s \succeq s_0$.*

Theorem 6.9 (Coarse-graining commutes with reconciliation up to controlled error). *Suppose the refinement square of Definition 6.8 is $(\varepsilon_{sr}^n, \varepsilon_{sr}^h)$-controlled. Define the coarse-stage consensus readout*

$$\mathcal{C}_r(y) := (n_r(y), h_r(y))$$

and the fine-then-coarse readout

$$\mathcal{C}_{s\to r}^{\mathrm{fine}}(x) := (\rho_{sr}(n_s(x)), \chi_{sr}(h_s(x))).$$

Then, for every fine state $x \in Q_s$, the product readout distance between $\mathcal{C}_{s\to r}^{\mathrm{fine}}(x)$ and $\mathcal{C}_r(\rho_{sr}(x))$ is bounded by

$$\max\{\varepsilon_{sr}^n, \varepsilon_{sr}^h\}.$$

Thus reconciling at the fine stage and then coarse-graining gives the same macroscopic law data as coarse-graining first and reconciling at the coarse stage, up to the declared control errors. If the exact naturality clauses of Definition 6.1 hold, then $\varepsilon_{sr}^n = \varepsilon_{sr}^h = 0$. If the errors are cofinally vanishing, the two procedures determine the same cylinder values in the inverse-limit topology at every fixed macroscopic stage.

Proof. The normal-form component of the product readout is bounded by the first inequality in Definition 6.8, and the obstruction component is bounded by the second. Taking the maximum gives the stated product bound. Exact naturality is precisely the special case

$$\rho_{sr} \circ n_s = n_r \circ \rho_{sr}, \qquad \chi_{sr} \circ h_s = h_r \circ \rho_{sr},$$

so both errors vanish there. Cofinal vanishing means that, for any fixed coarse cylinder and any desired readout tolerance, all sufficiently fine presentations give the same cylinder value within that tolerance, which is convergence in the inverse-limit topology. □

Remark 6.10 (What remains to verify on a concrete RG branch). Theorem 6.9 is not a claim that arbitrary renormalization maps commute with arbitrary repair laws. It identifies the exact branch burden: derive or estimate the two square defects ε_{sr}^{n} and ε_{sr}^{h} from the chosen coarse-graining channel, recovery map, decoder, and obstruction readout. The exact refinement theorem is the zero-defect case; approximate RG matching is theorem-grade only when these defects are explicitly controlled.

7 Record Algebras and the Operator Observation Layer

Inputs used here. From the fixed-cutoff collar package we use only the observer-accessible finite-dimensional algebra on one completed compare/write/verify slice, the declared pointer and overlap-sector projectors read on that same slice, and the trace-distance control on restored accessible states when restoration is invoked. No continuum lift and no broader observer-metaphysical assumption is used here.

Definition 7.1 (Exact record algebras and approximate record presentations)**.** *Fix one completed observer-accessible slice at cycle t, and let $\mathcal{A}_t^{\mathrm{acc}}$ be the corresponding finite-dimensional accessible algebra. An* exact record presentation *is a family of orthogonal projectors $\{\widehat{P}_a(t)\}_a \subset Z(\mathcal{A}_t^{\mathrm{acc}})$ whose generated algebra*

$$\mathcal{Z}_{\mathrm{rec}}(t) := \mathrm{Alg}(\{\widehat{P}_a(t)\}_a)$$

is finite and commutative.

An approximate record presentation *on the same declared readout slots is a family of projectors $\{P_a(t)\}_a \subset \mathcal{A}_t^{\mathrm{acc}}$ together with an exact record presentation $\{\widehat{P}_a(t)\}_a$ such that*

$$\delta_{\mathrm{rec}}(t) := \max_a \|P_a(t) - \widehat{P}_a(t)\|$$

is finite.

Theorem 7.2 (Record algebra, Born-Lüders update, and quantitative stability)**.** *Fix one completed observer-accessible slice at cycle t.*

1. *The exact record projectors generate a finite commutative central algebra $\mathcal{Z}_{\mathrm{rec}}(t) \subset Z(\mathcal{A}_t^{\mathrm{acc}})$.*

2. *For every event E in the finite event algebra generated by $\mathcal{Z}_{\mathrm{rec}}(t)$,*

$$\mathbb{P}_t(E) = \mathrm{Tr}(\rho_t \widehat{P}_E(t)), \qquad \rho_t|_E = \frac{\widehat{P}_E(t)\rho_t\widehat{P}_E(t)}{\mathrm{Tr}(\rho_t\widehat{P}_E(t))}.$$

3. *If no accepted repair between t and $t+1$ touches the support of $\widehat{P}_E(t)$, then the next read of E has probability 1.*

4. *If $\{P_a(t)\}_a$ is an approximate record presentation with modulus $\delta_{\mathrm{rec}}(t)$, then*

$$\|[P_a(t), P_b(t)]\| \le 4\,\delta_{\mathrm{rec}}(t)$$

for all declared record projectors $P_a(t), P_b(t)$. For every declared elementary record event a and every restored accessible state $\widetilde{\rho}_t$ satisfying

$$\|\widetilde{\rho}_t - \rho_t\|_1 \le \varepsilon,$$

one has

$$\left|\mathrm{Tr}(\widetilde{\rho}_t P_a(t)) - \mathrm{Tr}(\rho_t \widehat{P}_a(t))\right| \le \varepsilon + \delta_{\mathrm{rec}}(t).$$

Proof. Because the exact record projectors lie in the center of $\mathcal{A}_t^{\mathrm{acc}}$, they generate a finite commutative central algebra. Finite-dimensional projective measurement on that commuting algebra gives the Born trace and the Lüders conditioned state, proving (1) and (2).

If no accepted repair touches the support of $\widehat{P}_E(t)$, then the next completed read is performed on the same central projector surface. After conditioning on E, the state lies in the range of $\widehat{P}_E(t)$, so the next read of the same event has probability 1. This gives (3).

For (4), write

$$[P_a(t), P_b(t)] = [P_a(t) - \widehat{P}_a(t), P_b(t)] + [\widehat{P}_a(t), P_b(t) - \widehat{P}_b(t)],$$

because $[\widehat{P}_a(t), \widehat{P}_b(t)] = 0$. Since every projector has operator norm at most 1,

$$\|[P_a(t), P_b(t)]\| \le 2\|P_a(t) - \widehat{P}_a(t)\| + 2\|P_b(t) - \widehat{P}_b(t)\| \le 4\,\delta_{\mathrm{rec}}(t).$$

For the probability bound,

$$\left|\mathrm{Tr}\big(\widetilde{\rho}_t P_a(t)\big) - \mathrm{Tr}\big(\rho_t \widehat{P}_a(t)\big)\right| \le \left|\mathrm{Tr}\big((\widetilde{\rho}_t - \rho_t) P_a(t)\big)\right| + \left|\mathrm{Tr}\big(\rho_t (P_a(t) - \widehat{P}_a(t))\big)\right|.$$

The first term is bounded by $\|\widetilde{\rho}_t - \rho_t\|_1 \|P_a(t)\| \le \varepsilon$, and the second by $\|\rho_t\|_1 \|P_a(t) - \widehat{P}_a(t)\| \le \delta_{\mathrm{rec}}(t)$. Hence the total error is at most $\varepsilon + \delta_{\mathrm{rec}}(t)$. ☐

Merge boundary. What is closed here is the fixed-cutoff operator-algebraic observation surface: a central record algebra on the exact readout slice, Born/Lüders measurement on its event projectors, exact repeated-read stability under the untouched-support hypothesis, and explicit $(\varepsilon, \delta_{\mathrm{rec}})$ control when practical readout projectors are only close to that central surface. The broader export problem asks whether richer branches keep physically relevant pointer surfaces close to one such central record algebra and whether the same control survives refinement and continuation.

8 OPH Simulation Criterion and Fixed-Point Firewall

A fixed-point equation is necessary for a selected finite OPH packet, but it is not the definition of an OPH simulation. The additional clauses below are the nontrivial content.

Definition 8.1 (Nontrivial OPH simulation certificate)**.** *Work on the finite fixed-cutoff branch of Definition 5.20. Write*

$$Q := \Sigma/\Gamma, \qquad n := \overline{\mathrm{nf}}_\lambda : Q \to Q, \qquad N_\lambda := q(C).$$

Let $B : Q \to \mathcal{B}$ be a boundary/sector map preserved by accepted quotient repairs, fix $b \in \mathcal{B}$, and set

$$Q_b := B^{-1}(b), \qquad C_b := N_\lambda \cap Q_b.$$

For $u \in C_b$, let $\mathfrak{U}_u := \delta_u \in \Delta(Q)$. The selected pair (b, u), equivalently the packet $\mathfrak{U}_u$, has an ***OPH simulation certificate*** *when all of the following hold.*

(S1) ***Endogenous update.*** *The local maps are the recovery-derived collar maps of Definition 3.1, descend to the physical quotient, and use only the declared patch, overlap, recovery, decoder, and acceptance data. No undeclared oracle variable is an argument of an accepted physical update.*

(S2) ***Observer-readable records.*** *A declared observer-accessible region carries an exact record presentation* $\{\widehat{P}_a\}_a$ *as in Definition 7.1, with at least two nonzero orthogonal event projectors. Event probabilities, conditioned readouts, and any checkpoint/order relation interpreted as history are collected in a physical map*

$$\mathrm{Read} : Q \to \mathcal{Y}_{\mathrm{rec}}$$

whose terminal value depends only on the quotient normal form.

(S3) ***Overlap repair.*** *Every accepted nontrivial move strictly lowers the declared touched-overlap score, normal forms are exactly the globally consistent states, and the quotient normal form* n *is terminating and schedule-independent.*

(S4) ***Nontrivial branch elimination.*** *The selected fiber has one consistent quotient state but more than one candidate:*

$$C_b = \{u\}, \qquad Q_b \setminus C_b \neq \varnothing, \qquad n(Q_b) = \{u\}.$$

Thus at least one inconsistent candidate is genuinely removed instead of merely renamed a fixed point. A candidate with a nonzero declared holonomy or higher-gauge obstruction is rejected as obstructed and is not counted as an alternative selected world.

(S5) ***Implementation and clock closure.*** *The physical update and record readout have types*

$$n : Q \to Q, \qquad \mathrm{Read} : Q \to \mathcal{Y}_{\mathrm{rec}},$$

with no indispensable machine-state or external-time argument. More precisely, for any auxiliary implementation and clock sets $\mathcal{E}_{\mathrm{ext}}$ *and* Θ_{ext}*, let*

$$p : Q \times \mathcal{E}_{\mathrm{ext}} \times \Theta_{\mathrm{ext}} \to Q$$

be the physical projection. An admissible lifted update $\widetilde{n}$ *and readout* $\widetilde{\mathrm{Read}}$ *must satisfy*

$$p \circ \widetilde{n} = n \circ p, \qquad \widetilde{\mathrm{Read}} = \mathrm{Read} \circ p.$$

The asynchronous schedule and repair-step counter are proof data, not physical clock coordinates. Any history claimed by the branch is read from records in the terminal quotient state instead of supplied by an external clock.

The identities

$$n(u) = u, \qquad \mathcal{C}_\lambda(\mathfrak{U}_u) = \mathfrak{U}_u$$

are therefore consequences of the certificate, not its definition.

Theorem 8.2 (Selected OPH packet and fixed-point firewall)**.** *Assume Definitions 3.1, 3.2, and 3.3, Assumption 3.12, and the quotient and record hypotheses of Theorems 5.2, 5.17, and 7.2. Fix a preserved boundary/sector value* b_{OPH} *for which*

$$C_{b_{\mathrm{OPH}}} = \{u_{\mathrm{OPH}}\}, \qquad Q_{b_{\mathrm{OPH}}} \setminus C_{b_{\mathrm{OPH}}} \neq \varnothing,$$

and assume the selected observer surface has a nontrivial exact record presentation whose checkpoint/order data, when interpreted as history, are quotient-observable. Then the selected finite OPH packet

$$\mathfrak{U}_{\mathrm{OPH}} := \delta_{u_{\mathrm{OPH}}}$$

has an OPH simulation certificate in the sense of Definition 8.1. In particular,

$$n(u_{\mathrm{OPH}}) = u_{\mathrm{OPH}}, \qquad \mathcal{C}_\lambda(\mathfrak{U}_{\mathrm{OPH}}) = \mathfrak{U}_{\mathrm{OPH}}.$$

The converse is false: a fixed point, equilibrium, stationary point, or variational solution does not by itself imply an OPH simulation certificate. Indeed, for any set X with $|X| > 1$, the identity map $F = \mathrm{id}_X$ satisfies $F(x) = x$ for every $x \in X$, and every x is a global minimizer and stationary point of the constant functional $V \equiv 0$. Nevertheless, $F^{-1}(x) = \{x\}$, so no proper candidate basin collapses to x; clause (S4) *fails. The fixed-point or variational equation alone also supplies none of clauses* (S1)–(S3) *or* (S5)*.*

Proof. Clause *(S1)* is Definition 3.1 together with quotient descent from Theorem 5.2. Clause *(S2)* follows from Theorem 7.2; terminal representative independence is Theorem 5.17. Definition 3.2, Proposition 3.18, Proposition 3.21, Assumption 3.12, and Theorem 3.22 give clause *(S3)*.

For $x \in Q_{b_{\mathrm{OPH}}}$, boundary preservation gives $B(n(x)) = b_{\mathrm{OPH}}$, while repair completeness gives $n(x) \in N_\lambda$. Hence $n(x) \in C_{b_{\mathrm{OPH}}} = \{u_{\mathrm{OPH}}\}$, proving $n(Q_{b_{\mathrm{OPH}}}) = \{u_{\mathrm{OPH}}\}$. The assumed nonempty difference $Q_{b_{\mathrm{OPH}}} \setminus C_{b_{\mathrm{OPH}}}$ makes this elimination nontrivial. The holonomy and higher-gauge rejection statement is Theorems 4.1 and 4.4. This proves clause *(S4)*. Its proper-basin requirement is nonvacuous: on the verified rooted-tree packet domain of Theorem 3.14, fixing the root packet gives a singleton consistent fiber, while changing any non-root packet supplies an inconsistent candidate in that fiber.

The quotient normal-form and physical-record maps are defined on the declared physical state alone. Schedule independence removes the scheduler from the output, and Corollary 5.19 proves invariance under inert carrier enlargement. The displayed projection identities are the admissibility condition for any more general carrier or clock realization. Thus an auxiliary implementation state or iteration counter cannot change the physical result, proving clause *(S5)*. Finally, Theorem 5.21 gives

$$\mathcal{C}_\lambda(\delta_{u_{\mathrm{OPH}}}) = \delta_{n(u_{\mathrm{OPH}})} = \delta_{u_{\mathrm{OPH}}},$$

because $u_{\mathrm{OPH}} \in N_\lambda$. The identity-map and constant-functional example proves the final non-implication. □

Remark 8.3 (Structural falsification hooks)*.* An asserted OPH simulation certificate fails if any one of the following occurs: an accepted update depends on undeclared oracle, carrier, or external-clock data; the observer record algebra is trivial, unstable, or not quotient-readable; an accepted move fails strict mismatch descent; terminal states are not exactly the overlap-consistent states; two admissible schedules or two candidates in the selected boundary fiber produce different physical normal forms; the selected fiber has no inconsistent candidate to eliminate; a claimed global branch has nonzero holonomy or higher-gauge obstruction; or a carrier/clock lift changes the physical update or readout after projection. These are theorem-level failure conditions, not semantic disagreements about the word "simulation."

Remark 8.4 (Scope)*.* Theorem 8.2 certifies the named finite fixed-cutoff branch and selected boundary/sector fiber. It does not promote the finite packet closure to an arbitrary habitat-level state-and-law space, and it does not remove the branch and record hypotheses stated in the theorem.

9 Distributed Presentations of One Finite Universe

The implementation closure clause in Definition 8.1 has a distributed form. A worker partition is allowed to accelerate or package the computation, but it is not a new physical ingredient. The physical question is whether the worker run presents the same finite quotient repair system as the monolithic carrier.

Definition 9.1 (Finite global OPH carrier)**.** *At refinement stage r, a finite global OPH carrier consists of a finite patch graph*

$$G_r = (V_r, E_r),$$

finite patch state sets S_i, interface alphabets I_e, endpoint maps $\pi_{i,e} : S_i \to I_e$, an implementation-hiding group Γ_r, a boundary/sector map B_r, a mismatch potential Φ_r, accepted quotient repairs $\to_r$, and an observer-readable map

$$\mathrm{Read}_r : Q_r \to \mathcal{Y}_r, \qquad Q_r := \left(\prod_{i \in V_r} S_i\right) / \Gamma_r.$$

Let $C_r \subseteq Q_r$ be the quotient consistency set. On the finite branch covered by Theorems 3.22 and 5.2, the accepted quotient repair relation has a unique normal-form map

$$n_r : Q_r \to C_r.$$

A one-universe input is the pair $\mathbf{U}_r = (\mathfrak{U}_r, q_0)$, where $\mathfrak{U}_r$ is the carrier data above and $q_0 \in Q_r$. A worker partition is not part of $\mathfrak{U}_r$.

Definition 9.2 (Worker presentation and physical projection)**.** *Let $\kappa : V_r \to W$ assign every patch to one authoritative worker. The cut set is*

$$E_\kappa^{\mathrm{cut}} := \{\{i, j\} \in E_r : \kappa(i) \neq \kappa(j)\}.$$

A distributed presentation for κ has worker-owned states, ghost or halo copies, message queues, transaction records, retries, worker identifiers, checkpoints, event logs, and scheduler state. Let $\widetilde{Q}_{r,\kappa}$ be its quotient by purely local worker bookkeeping that leaves authoritative physical states unchanged. The physical projection

$$p_{r,\kappa} : \widetilde{Q}_{r,\kappa} \to Q_r$$

keeps only the authoritative patch states, quotients by Γ_r, and discards queues, retry metadata, worker labels, and external clocks. Let

$$D_r := \bigsqcup_\kappa \{\kappa\} \times \widetilde{Q}_{r,\kappa}, \qquad P_r(\kappa, \tilde{q}) := p_{r,\kappa}(\tilde{q}).$$

A distributed readout is physical when it factors as

$$\widetilde{\mathrm{Read}}_r = \mathrm{Read}_r \circ P_r.$$

Definition 9.3 (Admissible distributed event)**.** *For $d, d' \in D_r$, a distributed event $d \Rightarrow d'$ is **admissible** when one of the following holds.*

(a) ***Linearizable physical commit.*** *There is a nonempty monolithic accepted repair path*

$$P_r(d) \to_r^* P_r(d')$$

whose repair word is recorded as the event's linearization witness.

(b) ***Physical stutter.*** *$P_r(d') = P_r(d)$. Message delivery, halo refresh, prepare, abort, checkpoint, worker start or stop, idempotent replay, and repartition metadata are allowed only in this class unless they also carry a physical commit witness.*

(c) ***Certified rollback.*** *There is an earlier committed state d_j in the same run history such that $P_r(d') = P_r(d_j)$, and the rollback certificate names that committed projection root. Transparent restart from a committed frontier is a stutter.*

Proposition 9.4 (Normal form is constant on an accepted reachability cone)**.** *If $q \to_r^* q'$, then*

$$n_r(q') = n_r(q).$$

Proof. The state q' is reachable from q. The state $n_r(q')$ is a normal form reachable from q', hence also reachable from q. Theorem 3.22 and Theorem 5.2 give the unique quotient normal form reachable from q, so $n_r(q') = n_r(q)$. □

Theorem 9.5 (Distributed realization of one finite OPH universe)**.** *Fix a finite global carrier $\mathbf{U}_r = (\mathfrak{U}_r, q_0)$ satisfying Theorems 3.22 and 5.2. Let*

$$d_0 \Rightarrow d_1 \Rightarrow \cdots \Rightarrow d_m$$

be a finite distributed execution in D_r with $P_r(d_0) = q_0$, and assume every event is admissible in the sense of Definition 9.3. Then, for every t,

$$q_0 \to_r^* P_r(d_t), \qquad n_r(P_r(d_t)) = n_r(q_0).$$

If the final projection is a monolithic normal form, then

$$P_r(d_m) = n_r(q_0).$$

For every physical observable or observer readout $M : Q_r \to Y$,

$$M(P_r(d_m)) = M(n_r(q_0))$$

whenever $P_r(d_m)$ is normal. In particular, $\widetilde{\mathrm{Read}}_r(d_m) = \mathrm{Read}_r(n_r(q_0))$ for every distributed readout that factors through P_r.

Proof. Induct on t. The claim is true at $t = 0$. Suppose it holds at t. For a linearizable physical commit, admissibility gives

$$P_r(d_t) \to_r^* P_r(d_{t+1}),$$

so $q_0 \to_r^* P_r(d_{t+1})$, and Lemma 9.4 gives

$$n_r(P_r(d_{t+1})) = n_r(P_r(d_t)) = n_r(q_0).$$

For a physical stutter, $P_r(d_{t+1}) = P_r(d_t)$, so both statements are unchanged. For a certified rollback, $P_r(d_{t+1})$ is the projection of an earlier committed state; the induction hypothesis gives reachability from q_0 and equality of normal forms for that projection. This proves the two displayed identities for every t.

If $P_r(d_m)$ is a monolithic normal form, then it is the unique quotient normal form reachable from q_0, hence $P_r(d_m) = n_r(q_0)$. The observable and readout statements follow by applying M or Read_r to this equality and using $\widetilde{\mathrm{Read}}_r = \mathrm{Read}_r \circ P_r$. □

Corollary 9.6 (Partition, schedule, and restart invariance)**.** *Any two admissible distributed executions of the same finite carrier $\mathbf{U}_r$, with the same initial quotient state q_0, have the same terminal quotient observables and observer-readable outputs once their final projections are monolithic normal forms. The statement is independent of partition κ, worker count, scheduler choices, restart history, and repartition metadata.*

Proof. Apply Theorem 9.5 to each execution. Both final projections equal $n_r(q_0)$, so every quotient observable and every readout factoring through the projection has the same value. □

Proposition 9.7 (Restart stabilization separates safety from liveness)**.** *Assume Q_r is finite, every physical commit strictly descends the accepted potential until normality, every rollback returns to an earlier committed projection, only finitely many progress-erasing rollbacks occur, and after the last such rollback the scheduler is fair enough to commit an enabled repair whenever the projected state is not normal. Then the distributed run reaches a monolithic normal form after finitely many physical commits.*

Proof. Safety is Theorem 9.5. For liveness, ignore stutters and the finitely many progress-erasing rollbacks before the last one. After that time, each nonnormal projected state takes a strict descending accepted repair after finitely many scheduler steps. The set of possible potential values below the post-rollback value is finite, so strict descent can occur only finitely many times before a normal projected state is reached. □

Theorem 9.8 (Distributed refinement cube)**.** *Let $r \preceq s$. Suppose $\rho_{sr} : Q_s \to Q_r$ is a repair morphism, the corresponding holonomy maps form a cochain morphism, and distributed presentations at both stages are exact in the sense of Theorem 9.5. If the lifted restriction $\widetilde{\rho}_{sr} : D_s \to D_r$ commutes with physical projection,*

$$P_r \circ \widetilde{\rho}_{sr} = \rho_{sr} \circ P_s,$$

then distributed execution and coarse restriction commute on normal forms and readouts:

$$P_r(\widetilde{\rho}_{sr}(d^s_m)) = n_r(\rho_{sr}(q^s_0)) = \rho_{sr}(n_s(q^s_0))$$

whenever the fine final projection is normal. The same statement holds for holonomy readouts via χ_{sr}.

Proof. The fine distributed theorem gives $P_s(d^s_m) = n_s(q^s_0)$. Projection compatibility gives

$$P_r(\widetilde{\rho}_{sr}(d^s_m)) = \rho_{sr}(n_s(q^s_0)).$$

Theorem 6.5 identifies this value with $n_r(\rho_{sr}(q^s_0))$. The holonomy statement is exactly the cochain-morphism identity of Definition 6.6. □

Public certificate contract. A run pack that claims Theorem 9.5 must emit enough evidence for a verifier to reconstruct the premises without trusting worker narration. The required fields are: a global carrier manifest and hash, the monolithic graph G_r, the global initial quotient state q_0, a partition map κ, cut-interface records E^{cut}_κ with restriction maps, a global observer registry, code/config/run hashes, a committed event log with a linearization witness for each physical commit, stutter records for noncommitting worker events, rollback records naming earlier committed projection roots, repartition records preserving the physical projection root, a final monolithic normal-form certificate, and a final readout recomputed from the projected state. Missing, stale, shard-local, or synthetic replacements for these fields fail closed. The exact branch uses this contract; noisy branches additionally need the fair-block constants of Theorem 11.1.

10 Quotient Chart Transport and Neutral Geometry

The distributed certificate above proves that a worker presentation is one finite quotient system. A separate step is needed before observer rows from different shards may be read as a common neutral geometry.

Definition 10.1 (Common quotient chart atlas)**.** *For a shard s, let Σ_s be raw records, let Γ_s be the groupoid of declared presentation-only moves such as gauge representative changes and local port relabelings, and set*

$$\overline{Q}_s := \Sigma_s/\Gamma_s, \qquad X_s := n_s(\overline{Q}_s),$$

where n_s is the schedule-independent normal-form map. An interface atlas is a family of domains $U_{st} \subseteq X_s$, $U_{ts} \subseteq X_t$, and bijections

$$\tau_{ts} : U_{st} \to U_{ts}$$

satisfying identity, inverse, and cocycle laws, with zero closed-path holonomy on graph-shaped systems. A channel registry assigns each channel c a metric space (Z_c, d_c), weight $w_c > 0$, and local features $F_{s,c} : D_{s,c} \subseteq X_s \to Z_c$. The channel descends through the atlas when both domain membership and values are transported:

$$x \in D_{s,c} \Longleftrightarrow \tau_{ts}x \in D_{t,c}, \qquad F_{t,c}(\tau_{ts}x) = F_{s,c}(x).$$

Theorem 10.2 (Quotient-visible neutral readout)**.** *Under Definition 10.1, the relation generated by interface transport on $\bigsqcup_s X_s$ is an equivalence relation and defines*

$$Q_{\mathrm{vis}} := \left(\bigsqcup_s X_s\right) / \!\sim_\tau .$$

The canonical maps $X_s \to Q_{\mathrm{vis}}$ agree with transport, and zero closed-path holonomy makes them injective. Every compatible family of local channel maps descends uniquely to partial maps $F_c : Q_{\mathrm{vis}} \dashrightarrow Z_c$. Therefore, on the complete channel domain

$$Q_\star = \{x : F_c(x) \text{ exists for every } c \in \mathcal{C}_\star\},$$

the product formula

$$d_{\mathrm{neu}}(x, y) = \left(\sum_{c\in\mathcal{C}_\star} w_c\, d_c(F_c(x), F_c(y))^p\right)^{1/p}$$

is a well-defined pseudometric. It is a metric only after quotienting by zero-distance feature collisions, or after proving that the declared channel family separates points.

Proof. Identity, inverse, and cocycle laws give reflexivity, symmetry, and transitivity. The transport compatibility of local features is exactly the universal-property condition for descent through the quotient. The weighted product distance is then independent of the representative. Its triangle inequality is Minkowski's inequality applied to the channel metrics. The only possible failure of identity of indiscernibles is equality of all declared feature values, which is removed by quotienting feature-collision classes or by a joint-separation proof. □

Certificate boundary. Pairwise available-channel comparison is not a metric policy: different pairs can share different channels and violate the triangle inequality. A neutral-geometry run must therefore use complete cases, a fixed missing symbol whose mask is quotient-visible, or train-only imputation labelled as an imputed-representation metric. Presentation invariance requires a bijection of Q_{vis} and channel isometries, so gauge changes, port relabelings, observer order, repair schedule, and shard partition changes are checked on distance matrices, not on displayed coordinates. Refinement requires a cofinally vanishing tail modulus for the finite-stage distances. Euclidean claims require the double-centered Gram test

$$B = -\frac{1}{2}H(D^{\circ 2})H \succeq 0$$

and noisy runs report negative spectral mass, rank/effective rank, held-out stress, and positive/negative controls. Statistical certificates split independent generative batches before preprocessing; chart alignment, scaling, imputation, weights, graph construction, dimension selection, and thresholds are train/validation objects. Any shared shard batch, seed, boundary condition, trajectory family, duplicate, descendant, or repeated test-set inspection blocks the held-out theorem. This section certifies a common quotient metric or pseudometric only; physical Riemannian or Lorentzian spacetime identification belongs to the separate modular/geometric branch.

11 Noisy Fair-Block Approximate Consensus

The exact consensus theorem supplies the quotient normal-form target. A noisy implementation needs one more quantitative certificate: complete fair blocks must contract expected distance to that target. This section records the conditional bridge from local noisy repair to long-run observer-facing approximate consensus.

Theorem 11.1 (Global noisy approximate consensus under fair-block contraction)**.** *Let (Q, d_Q) be the observer-facing quotient state space of a fixed exported OPH patch federation, and let $\mathcal{N} \subset Q$ be the exact quotient normal-form set supplied by the finite repair theorem on that quotient. Define*

$$D(q) := d_Q(q, \mathcal{N}) = \inf_{n \in \mathcal{N}} d_Q(q, n).$$

Let $q_{t+1} = \widetilde{T}_{i_t,t}(q_t)$ be a noisy asynchronous repair process adapted to a filtration $(\mathcal{F}_t)_t$. Assume:

(G1) ***Exact quotient target.*** *The ideal repair relation on Q has the exact OPH normal-form package: finite exact descent, atomic conflict-component confluence, and repair completeness, so every fixed initial quotient state has a schedule-independent exact normal form in $\mathcal{N}$.*

(G2) ***Fair asynchronous blocks.*** *There are stopping times*

$$0 = \tau_0 < \tau_1 < \tau_2 < \cdots$$

such that each interval $[\tau_m, \tau_{m+1})$ contains enough local repairs to expose and act on every active mismatch class required by the exact transaction/confluence and repair-completeness certificate, with uniformly bounded length $\tau_{m+1} - \tau_m \le L$. Write the noisy block map as

$$\widetilde{B}_m := \widetilde{T}_{i_{\tau_{m+1}-1}, \tau_{m+1}-1} \circ \cdots \circ \widetilde{T}_{i_{\tau_m}, \tau_m}.$$

(G3) ***Uniform block contraction toward normal form.*** *There are constants $0 < \lambda < 1$ and $\varepsilon \ge 0$ such that, for every block m and every reachable quotient state q at time τ_m,*

$$\mathbb{E}\Big[D\Big(\widetilde{B}_m(q)\Big) \mid \mathcal{F}_{\tau_m}\Big] \le \lambda D(q) + \varepsilon.$$

(G4) ***Controlled within-block excursions.*** *There are constants $A \geq 1$ and $\beta \geq 0$ such that, for every $t \in [\tau_m, \tau_{m+1})$,*

$$\mathbb{E}[D(q_t) \mid \mathcal{F}_{\tau_m}] \leq A\, D(q_{\tau_m}) + \beta.$$

Then, at fair-block times,

$$\mathbb{E}[D(q_{\tau_m})] \leq \lambda^m D(q_0) + \frac{1 - \lambda^m}{1 - \lambda}\, \varepsilon,$$

and hence

$$\limsup_{m \to \infty} \mathbb{E}[D(q_{\tau_m})] \leq \frac{\varepsilon}{1 - \lambda}.$$

At all intermediate asynchronous times,

$$\limsup_{t \to \infty} \mathbb{E}[D(q_t)] \leq A\, \frac{\varepsilon}{1 - \lambda} + \beta.$$

Thus the noisy asynchronous OPH repair process converges in expectation to a controlled tube around the exact quotient normal-form set. If $\varepsilon = \beta = 0$, then $D(q_t) \to 0$ in L^1, hence also in probability, along the full asynchronous run.

Proof. Let $D_m := D(q_{\tau_m})$. Assumption (G3), applied to the realized state q_{τ_m}, gives

$$\mathbb{E}[D_{m+1} \mid \mathcal{F}_{\tau_m}] \leq \lambda D_m + \varepsilon.$$

Taking expectations,

$$\mathbb{E}[D_{m+1}] \leq \lambda\, \mathbb{E}[D_m] + \varepsilon.$$

Iterating the scalar recursion gives

$$\mathbb{E}[D_m] \leq \lambda^m D(q_0) + \varepsilon \sum_{r=0}^{m-1} \lambda^r = \lambda^m D(q_0) + \frac{1 - \lambda^m}{1 - \lambda} \varepsilon.$$

Taking $m \to \infty$ yields the fair-block limsup bound. For $t \in [\tau_m, \tau_{m+1})$, Assumption (G4) gives

$$\mathbb{E}[D(q_t)] \leq A\, \mathbb{E}[D(q_{\tau_m})] + \beta.$$

Substitution of the fair-block estimate and then taking the limsup over all intermediate times gives

$$\limsup_{t \to \infty} \mathbb{E}[D(q_t)] \leq A\, \frac{\varepsilon}{1 - \lambda} + \beta.$$

If $\varepsilon = \beta = 0$, then $\mathbb{E}[D(q_{\tau_m})] \leq \lambda^m D(q_0) \to 0$. Since $D \geq 0$, convergence in expectation to zero implies convergence in probability at block times. Assumption (G4) then gives $\mathbb{E}[D(q_t)] \leq A\, \mathbb{E}[D(q_{\tau_m})]$ inside each block, so the same L^1 and probability convergence holds along the full asynchronous run. ☐

Corollary 11.2 (Approximate observer-facing schedule independence)**.** *Let $M : Q \to Y$ be an observer-facing readout into a metric space (Y, d_Y), and suppose M is L_M-Lipschitz. Fix an exact sector ζ whose exact normal-form set is the singleton $\mathcal{N}_\zeta = \{n_\zeta\}$, and apply Theorem 11.1 on that sector, so that $D(q) = d_Q(q, n_\zeta)$. Then*

$$\limsup_{t \to \infty} \mathbb{E}[d_Y(M(q_t), M(n_\zeta))] \leq L_M \left(A\, \frac{\varepsilon}{1 - \lambda} + \beta \right).$$

For two noisy asynchronous schedules q_t and q'_t started in the same exact singleton sector and satisfying the same certificate,

$$\limsup_{t \to \infty} \mathbb{E}\big[d_Y(M(q_t), M(q'_t))\big] \leq 2L_M \left(A\, \frac{\varepsilon}{1 - \lambda} + \beta \right).$$

Proof. The Lipschitz condition gives

$$d_Y(M(q_t), M(n_\zeta)) \le L_M d_Q(q_t, n_\zeta) = L_M D(q_t).$$

The first bound follows from Theorem 11.1. The two-schedule bound follows from the triangle inequality through $M(n_\zeta)$ and applying the first estimate to both schedules. ☐

Corollary 11.3 (High-probability noisy consensus tube)**.** *Assume the block-distance process also admits the pathwise decomposition*

$$D_{m+1} \le \lambda D_m + \varepsilon + \xi_{m+1}, \qquad \mathbb{E}[\xi_{m+1} \mid \mathcal{F}_{\tau_m}] = 0, \qquad |\xi_{m+1}| \le b$$

almost surely. Then, for every $a > 0$,

$$\Pr\left[D_m > \lambda^m D_0 + \frac{1-\lambda^m}{1-\lambda}\varepsilon + a\right] \le \exp\left(-\frac{a^2(1-\lambda^2)}{2b^2}\right).$$

Proof. Unrolling the recursion gives

$$D_m \le \lambda^m D_0 + \frac{1-\lambda^m}{1-\lambda}\varepsilon + \sum_{r=1}^{m} \lambda^{m-r}\xi_r.$$

The final term is a weighted martingale sum with increments bounded by $|\lambda^{m-r}\xi_r| \le \lambda^{m-r} b$. Azuma–Hoeffding gives

$$\Pr\left[\sum_{r=1}^{m} \lambda^{m-r}\xi_r > a\right] \le \exp\left(-\frac{a^2}{2\sum_{r=1}^{m}\lambda^{2(m-r)}b^2}\right).$$

Since $\sum_{r=1}^{m}\lambda^{2(m-r)} \le (1-\lambda^2)^{-1}$, substitution yields the claimed bound. ☐

Proposition 11.4 (Finite fair-block contraction certificate)**.** *Let Q be finite, let $\mathfrak{B}_{\mathrm{fair}}$ be a finite list of noisy fair-block types, and let $K_B(q, q')$ be the Markov kernel induced by block type B. If there are constants $0 < \lambda < 1$ and $\varepsilon \ge 0$ such that*

$$\sum_{q' \in Q} K_B(q, q') D(q') \le \lambda D(q) + \varepsilon \qquad \forall q \in Q, \quad \forall B \in \mathfrak{B}_{\mathrm{fair}},$$

then Assumption (G3) of Theorem 11.1 holds for any run whose fair blocks are drawn from $\mathfrak{B}_{\mathrm{fair}}$.

Proof. Conditioning on the current quotient state q and the realized fair-block type B, the conditional expectation of D after the block is exactly $\sum_{q'} K_B(q, q') D(q')$. The displayed inequality is therefore Assumption (G3), uniformly over all reachable states and fair-block types. ☐

Finite audit route and constants. For a finite exported packet net, Proposition 11.4 gives the practical certificate path:

finite quotient $Q \Rightarrow$ exact normal forms $\mathcal{N} \Rightarrow$ distance table $D(q) \Rightarrow$ noisy fair-block kernels $K_B \Rightarrow (\lambda, \varepsilon)$ certifica

Here $\mathcal{N}$ is the exact quotient normal-form set, $D(q)$ is observer-facing residual distance to that set, λ is the net contraction produced by one completed fair block, ε is the per-block irreducible local recovery / record / readout / calibration / environmental noise, A bounds transient within-block expansion, β is the within-block noise floor, and L is the fairness horizon before all required active mismatch classes are serviced. In quantum/collar implementations, ε may absorb Petz-domain truncation, Fawzi–Renner recovery error, approximate central-record error, detector/readout noise, and coarse-graining defect.

Claim boundary. Theorem 11.1 is a conditional global noisy-consensus theorem. It does not follow from the exact finite repair theorem alone. The exact OPH theorem supplies the quotient normal-form target $\mathcal{N}$; the noisy theorem requires a separate fair-block contraction certificate $(\lambda, \varepsilon, A, \beta, L)$ for the chosen implementation. Without that certificate, OPH retains exact fixed-cutoff convergence and the collar-local splice and record-stability estimates above. With that certificate, arbitrarily long asynchronous noisy repair sequences converge to a controlled observer-facing tube around the exact quotient normal-form set.

12 Law-Space Selection and Observer Emergence

This section studies a simple meta-selection model on law space. The aim is to formalize one criterion for favoring schedule-robust, observer-supporting, and simple laws; the replicator dynamics below is part of the model, not a claim about literal cosmological dynamics.

We begin by defining what it means for a law to support observers. An observer is treated operationally as a **persistent predictive module**: a subgraph that maintains a stable record algebra and uses its output law to predict its boundary's future behavior.

Definition 12.1 (Schedule robustness)**.** *Fix distributions μ over initial conditions and ν over asynchronous schedules, and a gauge-invariant observable M. For a law λ, define*

$$\mathcal{R}_M(\lambda) := \Pr_{s\sim\mu,\, \sigma,\tau\sim\nu}\left[M\big(\mathrm{nf}_\lambda^\sigma(s)\big) = M\big(\mathrm{nf}_\lambda^\tau(s)\big)\right].$$

If Theorem 3.22 holds for λ, then $\mathcal{R}_M(\lambda) = 1$.

Definition 12.2 (Observer yield)**.** *Let X_t^λ denote the stationary process obtained by repeated local perturbation plus reconciliation under law λ. For each subgraph $U \subseteq V$, let $\mathcal{Z}_U^{\mathrm{rec}}(t)$ be the declared exact record algebra on its observer-accessible surface, or the reference exact algebra when only an approximate record presentation is available, and let $Y_U(t)$ be the corresponding finite outcome variable induced by that record algebra. Then U is (η, ε, h)-observer-like if it is record-stable:*

$$d_{\mathrm{TV}}\big(\mathrm{Law}(Y_U(t+h)), \mathrm{Law}(Y_U(t))\big) \le \eta,$$

and predictive:

$$I\big(Y_U(t);\; X^\lambda_{\partial U,\, t+1:t+h}\big) \ge \varepsilon.$$

Define

$$\mathcal{O}_{\eta,\varepsilon,h}(\lambda) := \mathbb{E}[\#\{U \subseteq V : U \text{ is } (\eta,\varepsilon,h)\text{-observer-like}\}].$$

Definition 12.3 (Law fitness)**.** *Let $K(\lambda)$ be a description-length penalty. Define*

$$f(\lambda) = \alpha\, \mathcal{R}_M(\lambda) + \beta\, \mathcal{O}_{\eta,\varepsilon,h}(\lambda) - \gamma\, K(\lambda),$$

with $\alpha, \beta, \gamma > 0$.

Theorem 12.4 (Replicator monotonicity on law space)**.** *Let $\Lambda = \{\lambda_1, \dots, \lambda_m\}$ be candidate laws with population weights $x_i(t)$ under replicator dynamics:*

$$\dot{x}_i = x_i(f_i - \bar{f}), \qquad f_i := f(\lambda_i), \qquad \bar{f} := \sum_{j=1}^m x_j f_j.$$

Then

$$\frac{d}{dt}\bar{f} = \mathrm{Var}_x(f) \ge 0.$$

Mean fitness is nondecreasing, and strictly increasing unless all extant laws have the same fitness.

Proof. Direct computation:

$$\frac{d}{dt}\bar{f} = \sum_i \dot{x}_i f_i = \sum_i x_i(f_i - \bar{f})f_i = \sum_i x_i f_i^2 - \bar{f}^2 = \mathrm{Var}_x(f) \geq 0.$$

Equality iff all f_i on the support of x are equal. □

This theorem records the monotonicity property of the meta-selection model.

13 Connection to Observer-Patch Holography

The formalism above is the computational skeleton of Observer-Patch Holography (OPH). The observer patches carry von Neumann algebras on support-visible holographic cuts. In symmetric regulator charts those cuts may be represented by patches on a screen S^2. The fixed-cutoff microphysics carrier is a federated patch system with echosahedral local interfaces. The S^2 chart supplies cap and collar geometry, and in the companion relativity branch its conformal group supplies the Lorentz bridge. The carrier supplies finite ports, records, and repair interfaces. The overlap projections are restrictions to shared subalgebras, and the consistency condition is algebraic state agreement on overlaps.

The bridge to physics works as follows in the broader companion corpus:

- The patch net becomes a net of support-visible subregion algebras; S^2 is the standard observer-facing regulator chart, not a required literal substrate.
- Overlap Consistency, one of the canonical framework axioms, is the algebraic version of Definition 2.1.
- The recoverability clause of the canonical Recoverable Generalized Entropy axiom provides controlled collar recovery data. The exact collar factorization $\rho_{ABD} = \bigoplus_\alpha p_\alpha \, \rho^{(\alpha)}_{Ab^\alpha_L} \otimes \rho^{(\alpha)}_{b^\alpha_R D}$ is used only at exact Markovity *together with* the Markov-split alignment of the HJPW factors with the preselected edge split (the compact paper's alignment hypothesis; exact Markovity alone yields the normal form only over a state-dependent split), or along the fixed-collar replacement limit with EC-aligned comparison states, while the declared Petz/Fawzi–Renner recovery channels supply recovered comparison states from which the local repair moves are built (see Appendix A and Definition C.3).
- Gauge symmetry as implementation hiding (Theorem 5.2) becomes the fixed-cutoff edge-sector seed package. On the companion compact paper's branch carrying the explicit compact-gauge refinement receipt, coherent surjective pullback functors and compatible forgetful fibers reconstruct a compact group from the tensor-generated combined-zero-obstruction sectors. Here zero obstruction means central or higher-associator strictifiability together with at least one allowed strict 1-cocycle representative having trivial represented holonomy. This supplies receipt-conditional transport across cutoffs and classification. The cofinal realized witness uses the same receipt, and MAR plus the explicit one-Higgs matter package then selects the Standard Model quotient $\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)/\mathbb{Z}_6$ together with the exact hypercharge lattice and the realized counts $N_c = 3$, $N_g = 3$.
- On the declared support-visible compact-gauge branch, the companion compact paper obtains the four-dimensional Euclidean Yang–Mills form from compact-gauge holonomy data and the local MaxEnt/Gibbs continuum limit only with the branch's renormalized four-dimensional

identification receipt. The compact paper separately proves projective weak-* / GNS extraction from its finite cylinder system. On a finite support quotient, weighted resampling inside observation fibers is the orthogonal conditional-expectation projector. A concrete active repair kernel has that status only when its independently extracted transition matrix satisfies the support, equal-fiber-row, and weighted detailed-balance receipt of Ref. [1]; constructing the tested matrix from the target projector formula is not a verification. Identification with Euclidean transfer, vacuum persistence, OS reconstruction, noncollapse, and passage of the uniform repair gap require their separately named finite and continuum receipts. On that certified branch the Yang–Mills gap equals the repair gap; coherent cylinder extraction alone does not make that identification or supply a Clay-admissible theory.

- The coarse-graining compatibility theorem (Theorem 6.9) is the link between the finite reconciliation protocol and the refinement/RG language used by the OPH branches: the macroscopic law space is stable when the selected coarse-graining channel shadows the normal-form and obstruction maps with controlled defects.
- Stable defects (Corollary 4.3) become the topologically protected excitations identified with particles on the declared branch.
- The record-algebra theorem (Theorem 7.2) provides the formal basis for the fixed-cutoff observation layer, where records are carried by central or quantitatively stable approximately commuting projectors in overlap centers.

The companion manuscripts develop a derived gravity branch from entanglement equilibrium and modular geometry, the SM gauge-group and count closure from edge-sector reconstruction plus the realized MAR branch, a conditional support-visible compact-gauge Yang–Mills form and repair-gap theorem under the separately named continuum-identification, transfer, OS/noncollapse, and uniform-gap receipts, and a controlled large-N_{edge} worldsheet effective-description branch above the heat-kernel edge-sector identity. The implemented cosmological capacity branch, with its self-closure target (readback map not yet constructed; CL-7), and later phenomenological continuations are kept explicitly separate from that recovered-core claim set.

This paper provides the finite patch-net foundation for the broader companion corpus.

14 Discussion and Scope Boundaries

The fixed-point consensus spine of OPH consists of a total, idempotent, boundary-preserving quotient repair operator on the declared finite branch; schedule-independent normal forms from fixed initial quotient states; boundary-conditioned uniqueness when a preserved boundary/sector fiber has a unique consistent extension; a finite layered carrier proving $H_B \wedge H_{\mathrm{fib}}$; nontrivial functional selected-fiber branch elimination; holonomy obstructions; gauge-quotient invariance, separated cofinal refinement-limit consensus classes, controlled coarse-graining compatibility, the fixed-cutoff operator-record theorem, distributed one-universe realization for admissible worker presentations of a single global carrier, and conditional noisy fair-block convergence once a separate contraction certificate is supplied. It also proves the full repair-completeness, Petz-domain, and quotient-compatibility package on the rooted-tree packet-net domain of Theorem 3.14, and gives a clean law-selection meta-model. The relativity chain and the realized Standard Model structural chain are the recovered core. The capacity relation is a separate implemented branch with a self-closure target (readback map not yet constructed; CL-7). Downstream phenomenology requires additional assumptions beyond the consensus results proved here.

Complexity boundary. On a fixed finite patch net, the accepted reconciliation dynamics is a finite-state asynchronous rewrite system on Σ. Under Theorem 3.22, every accepted repair run has at most the number of distinct reachable μ-values below its start value minus one, because each accepted transaction strictly lowers the declared exact measure. On the scalar Φ-branch this specializes to the older $|\Phi(\Sigma)| - 1 \leq |\Sigma| - 1$ bound. Exact normal-form computation is therefore decidable by direct iteration of accepted aggregate transactions. This step bound is a termination statement; the schedule-independent answer depends on the atomic conflict-component local diamond and repair-completeness clauses of Theorem 3.22. What this paper does *not* prove is a uniform polynomial-time bound, a sharper complexity-class placement for families of growing patch nets, or any hardness lower bound.

Approximate-stability boundary. The theorem-grade consensus statement is exact on the declared fixed-cutoff branch. Approximate control begins collar-locally: Theorem A.1 gives the exact-Markov modulus $\delta^{\mathrm{M}}_{A:B:D}(\varepsilon) \to 0$ on one fixed finite-dimensional collar model and the one-shot recovery comparison bound $2\sqrt{1 - e^{-\varepsilon}} \leq 2\sqrt{\varepsilon}$, while Theorem 7.2 gives the $(\varepsilon, \delta_{\mathrm{rec}})$ repeated-read stability bound for approximate record projectors. The long-run noisy statement is conditional: Theorem 11.1 upgrades those local noisy controls to a global expected tube around the exact quotient normal-form set only after a fair-block contraction certificate $(\lambda, \varepsilon, A, \beta, L)$ is supplied for the chosen implementation. It does not follow from finite descent or fairness alone, and it gives a unique approximate readout only inside singleton boundary/sector fibers. Ref. [1] supplies a complementary receipt-only endpoint estimate from a residual error-bound modulus and an inverse-observation modulus, together with refinement comparison bounds. Those results do not manufacture the fair-block contraction used here and do not imply pathwise long-run confinement under persistent noise.

Expressive-power boundary. The law-selection model of Theorem 12.4 is a finite-candidate monotonicity result. For each fixed patch net the theorem package proves a finite-state exact reconciliation mechanism. A universality claim would require an explicit uniform family of patch nets and repair laws that simulates arbitrary circuits or machines with stated encoding overhead and robustness under asynchronous schedules. No such theorem is supplied here.

With those boundaries explicit, the consensus-paper boundaries and companion interfaces are:

1. **Sharper RG-shadowing estimates.** Theorem 6.9 closes the abstract reconciliation/coarse-graining square once its normal-form and obstruction defects are supplied. The quantitative task is to derive model-specific or uniform bounds for ε^n_{sr} and ε^h_{sr} from concrete OPH coarse-graining channels and recovery decoders.

2. **Defect classification and refinement-limit transportability.** The fixed-cutoff hierarchy extends from abelian frustrations to crossed-module classes $q \in \check{H}^2(N, H \to G)$. The continuation task is to connect those higher-gauge defect sectors to the refinement-stable transportable sector category used in the broader compact-gauge reconstruction lane.

3. **Observable-level confluence beyond the declared fixed-cutoff physical algebra.** Theorem 5.17 closes the fixed-cutoff quantum-lift statement when microscopic representatives differ by gauge relabelings globally or by sector/higher-gauge relabelings on the same declared quotient-local glued state. The continuation question is whether an analogous observable theorem survives on broader refinement-stable branches where the union-collar compatibility is only approximate or where the physical observable algebra itself changes under refinement.

4. **Distributed implementation certificates.** Theorem 9.5 proves invariance for a worker presentation only after the implementation supplies one global carrier, authoritative ownership, cut interfaces, projection-preserving events, rollback roots, and final monolithic normal-form/readout certificates. Shard-local seeds, missing cut artifacts, stale manifests, synthetic seam trajectories, or worker-id histories do not certify a one-universe realization.

5. **Global approximate-consensus stability.** Theorem A.1 and Theorem 7.2 supply the collar-local perturbative controls carried by this paper. The rooted-tree packet domain proves the exact finite repair package on one nontrivial exported domain. Theorem 11.1 closes the long-run noisy branch only after a fair-block contraction certificate is supplied. What remains absent is an automatic theorem showing that arbitrary approximate recovery moves on arbitrary packet-closed exported overlap nets are repair-complete, preserve transactional validation and atomic conflict-component confluence, satisfy the support/CPTP clause on every Petz branch used there, and meet that fair-block contraction certificate.

6. **Expressive power / universality.** The paper supplies no uniform simulation construction, so universality is outside the theorem-grade output.

7. **Applied material, plasma, and device branches.** The fixed-point theorem classifies quotient normal forms and obstructions once a physical quotient and repair law are declared. It does not select a Hall sector, material order, confinement regime, nuclear yield, or hardware performance number without a quotient-intrinsic ensemble, source action, repair ledger, and evidence receipts for that branch.

8. **Interface to the companion gravity/gauge stack.** The companion compact paper derives the gravity chain through the support-visible BW scaling theorem on the prime geometric cap subnet. This consensus paper supplies the finite-state and refinement-consensus spine used by that bridge, while dark-sector proposals, heuristic baryogenesis continuations, spectroscopy, and string/worldsheet topics remain separate continuations that require their own declared inputs beyond what this paper itself proves.

The last item is an interface statement instead of an extra consensus premise. The fixed-point theorems proved here stand on their stated finite and refinement-consensus hypotheses, while companion gravity, gauge, and continuation sectors add their own theorem surfaces and declared inputs.

14.1 Assumption-Dependent BFT and QECC Extensions

The consensus formalism of OPH has natural analogies to classical and quantum distributed Byzantine agreement. Observer patches correspond to protocol nodes, overlap repair corresponds to a quorum vote, and the repair fixed-point corresponds to a consensus state. Under explicit structural assumptions (partial synchrony, one-vote-per-view, certificate semantics, authentication, and quorum overlap: either the classical exact sizing $n = 3f + 1, q = 2f + 1$ or a general threshold q with $2q \geq n + f + 1$), a QBFT-style interpretation of OPH repair satisfies safety and liveness (Appendix C, Theorem C.2). On the fixed-cutoff collar branch used here, the repair map is written in exact-splice / Petz form; the assumption-dependent item is the CPTP property on all inputs, which requires either full-rank $\mathcal{N}(\sigma)$ or an explicit domain restriction, and trace-preserving completion is not automatic when $\mathcal{N}(\sigma)$ has a non-trivial kernel (Proposition C.5). A quantum error-correcting interpretation is possible only after a genuine code subspace, logical dictionary, error family, and recovery map are supplied. The graph-min-cut equality for distance is not a property of a bare overlap graph: the same graph can realize distance 1 or distance $|V|$ under different interface maps (Theorem C.10). Distance/min-cut statements require the topological-code certificate of Definition C.11 and Theorem C.12; resilience requires the Knill–Laflamme certificate of Theorem C.14. All of these extensions are assumption-dependent or conjectural and are not part of the core theorem package of Paper 4.

Desired statement	**Required certificate**
Overlap network is a code	finite constraint-code data of Proposition 2.3
Local repair is a physical map	$\mathrm{locRep}_\lambda : Q \to Q$ on the finite quotient presentation, with boundary preservation and exact descent
Global repair is a physical normal-form map	$\mathrm{Rep}_\lambda = \overline{\mathrm{nf}}_\lambda : Q \to Q$, total, idempotent, schedule-independent, and landing in C_Q
Repair respects gauge	quotient-valued $\mathrm{Rep}^\Sigma_\lambda = \mathrm{Rep}_\lambda \circ q$, hence invariant under Γ
Boundary reconstruction	layered finite carrier with $H_B \wedge H_{\mathrm{fib}}$ and $\mathrm{Rep}_\lambda(x) = E(B(x))$ on admissible fibers
Repair converges	finite exact descent; confluence additionally needs atomic conflict-component local diamond and repair completeness
Distance equals min-cut	topological-code certificate with homological logicals and matching systole/min-cut geometry
Corrects t corrupted carriers	code projector, error family, Knill–Laflamme condition, and certified $t < d/2$ distance bound
Exponential convergence	declared transfer/channel operator with stationary projection and spectral gap
Long-run noisy approximate consensus	fair-block contraction certificate $(\lambda, \varepsilon, A, \beta, L)$ for distance to the exact quotient normal-form set
Wall-clock BFT liveness	partial synchrony, quorum certificates, authentication, and quorum-overlap sizing: $n = 3f+1, q = 2f+1$ or $2q \geq n+f+1$
Hardware search work reduction	exact-verifier candidate-enrichment factor measured under controls

A Quantum/Algebraic Lift: Markov-Collar Splice Theorem

This appendix records the algebraic splice statement used to relate the finite patch-net model to the OPH collar formalism.

An observer is written as

$$O = (P, \mathcal{A}(P), \rho, R),$$

where P is the screen patch, $\mathcal{A}(P)$ the local von Neumann algebra, ρ the local state, and R the record algebra.

Theorem A.1 (Markov-collar splice theorem, exact and controlled). *Suppose a collar tripartition A-B-D has the EC-aligned exact Markov decomposition*

$$\rho_{ABD} = \bigoplus_\alpha p_\alpha \, \rho^{(\alpha)}_{Ab^\alpha_L} \otimes \rho^{(\alpha)}_{b^\alpha_R D}$$

over the preselected edge factors. This displayed form is the hypothesis: by the compact paper's Markov-split alignment analysis it is strictly stronger than $I(A : D \mid B)_\rho = 0$, which by HJPW yields such a factorization only over a state-dependent split of B. Let $\sigma^{(\alpha)}_{b^\alpha_R D'}$ be any family of normalized environment states compatible with the same right-boundary sectors. Define

$$\rho'_{ABD'} = \bigoplus_\alpha p_\alpha \, \rho^{(\alpha)}_{Ab^\alpha_L} \otimes \sigma^{(\alpha)}_{b^\alpha_R D'}.$$

Then for every observable X supported on $A \cup b_L$,

$$\mathrm{Tr}(X\rho'_{ABD'}) = \mathrm{Tr}(X\rho_{ABD}).$$

Fix one finite-dimensional collar model and let

$$\mathfrak{M}_{A:B:D} := \{\tau_{ABD} : I(A:D \mid B)_\tau = 0\},$$

with exact-Markov distance modulus

$$\delta^{\mathrm{M}}_{A:B:D}(\varepsilon) := \sup\left\{\inf_{\tau \in \mathfrak{M}_{A:B:D}} \|\omega - \tau\|_1 : I(A:D \mid B)_\omega \leq \varepsilon\right\}.$$

Then

$$\delta^{\mathrm{M}}_{A:B:D}(\varepsilon) \to 0 \qquad (\varepsilon \downarrow 0).$$

On a fixed faithful collar class with lower floor $\lambda_* > 0$, this qualitative modulus can be sharpened to a collar-local rate

$$\delta^{\mathrm{M},\lambda_*}_{A:B:D}(\varepsilon) \leq C_{A:B:D,\lambda_*}\, \varepsilon^{\theta_{A:B:D,\lambda_*}},$$

by the compact real-analytic Lojasiewicz inequality applied to $I(A:D \mid B)$. The constants depend on the fixed collar model and floor; they are not a dimension-free stability theorem for arbitrary tripartite systems. Hence if $I(A:D \mid B)_\omega \leq \varepsilon$ and $\widetilde{\omega}_\varepsilon \in \mathfrak{M}_{A:B:D}$ is chosen so that

$$\|\omega - \widetilde{\omega}_\varepsilon\|_1 \leq \delta^{\mathrm{M}}_{A:B:D}(\varepsilon),$$

the corresponding exact splice $\widetilde{\omega}'_\varepsilon$ satisfies

$$\left|\mathrm{Tr}(X\omega) - \mathrm{Tr}(X\widetilde{\omega}'_\varepsilon)\right| \leq \|X\|_\infty\, \delta^{\mathrm{M}}_{A:B:D}(\varepsilon)$$

for every observable X supported on $A \cup b_L$.

Independently, if $I(A:D \mid B)_\omega \leq \varepsilon$, then there exists a recovery map $\mathcal{R}_{B\to BD}$ such that

$$\|\omega_{ABD} - (\mathrm{id}_A \otimes \mathcal{R}_{B\to BD})(\omega_{AB})\|_1 \leq 2\sqrt{1-e^{-\varepsilon}} \leq 2\sqrt{\varepsilon}.$$

Proof. The exact splice statement is the usual blockwise factorization argument:

$$\mathrm{Tr}(X\rho'_{ABD'}) = \sum_\alpha p_\alpha\, \mathrm{Tr}\Big(X\, \rho^{(\alpha)}_{Ab^\alpha_L}\Big)\, \mathrm{Tr}\Big(\sigma^{(\alpha)}_{b^\alpha_R D'}\Big).$$

Each right factor is normalized, so the value agrees with the same computation for ρ_{ABD}.

For the controlled statement, compactness of the fixed finite-dimensional state space and continuity of conditional mutual information imply $\delta^{\mathrm{M}}_{A:B:D}(\varepsilon) \to 0$: otherwise one could find a sequence with $I(A:D \mid B) \to 0$ staying a fixed trace distance away from every exact Markov state, contradicting convergence of a subsequence to an exact Markov limit point. The splice identity additionally requires $\widetilde{\omega}_\varepsilon$ to be EC-aligned over the preselected edge factors; small conditional mutual information supplies closeness only to the full exact Markov set, so the existence of EC-aligned replacements is carried as an explicit hypothesis on the controlled family (the compact paper's Markov-split alignment hypothesis). Once such an EC-aligned $\widetilde{\omega}_\varepsilon$ is chosen, the exact splice identity for $\widetilde{\omega}_\varepsilon$ gives

$$\left|\mathrm{Tr}(X\omega) - \mathrm{Tr}(X\widetilde{\omega}'_\varepsilon)\right| = |\mathrm{Tr}[X(\omega - \widetilde{\omega}_\varepsilon)]| \leq \|X\|_\infty\, \delta^{\mathrm{M}}_{A:B:D}(\varepsilon).$$

The final inequality is the standard Fawzi–Renner recovery bound [16]. □

This appendix therefore uses exact splice identities in only two regimes: literal EC-aligned exact Markovity, or a controlled collar family on one fixed finite-dimensional model for which $\delta^{\mathrm{M}}_{A:B:D}(\varepsilon) \to 0$ and EC-aligned replacements exist. Fawzi–Renner recovery supplies the constructive recovered comparison state; the fixed-collar modulus supplies the exact-Markov comparison. Small one-shot conditional mutual information is not silently upgraded to an exact normal form.

For later dark-sector continuations, the exact finite identity is only an expectation identity. For a faithful finite state ρ_{ABD}, with $K_X = -\log \rho_X$,

$$\operatorname{Tr} \rho_{ABD}(K_{AB} + K_{BD} - K_B - K_{ABD}) = I_\rho(A : D \mid B).$$

This does not identify raw CMI with a state-independent local stress source. Near a full-rank exact Markov state σ, a perturbation $\rho(t) = \sigma + tX + O(t^2)$ has

$$\frac{\mathrm{d}}{\mathrm{d}t} I_{\rho(t)}(A : D \mid B)\bigg|_{t=0} = 0,$$

while a fixed-reference modular-energy variation is generically linear in t. Recovery bounds therefore do not by themselves prove a local stress-source theorem; source-specific coarse graining, collar localization, cover independence, and finite-model proof receipts are separate continuation data.

B Fixed-Cutoff Realization, Quotient Repair, and Edge Centers

This appendix carries the fixed-cutoff realization and edge-center items for the consensus paper. They sharpen the quotient-first repair interpretation used throughout the consensus paper and make the collar boundary data explicit at the same finite patch-net level. On the declared fixed-cutoff collar branch, the local repair step is read from exact Markov splice or a declared Petz/Fawzi–Renner recovery move on that same collar data; the recovery move gives a recovered comparison state, while exact splice requires exact Markovity or a controlled fixed-collar replacement modulus. Representative repair maps are only lifts of the resulting quotient-local update.

B.1 Quotient Repair and UV Underdetermination

At fixed cutoff, each regulator cell x carries a finite-dimensional factor $\mathfrak{h}_x$, patch algebras are finite type-I algebras, and gauge-as-gluing is realized as a compact boundary redundancy action on cut data. The physical repair law therefore belongs on the overlap-invariant quotient rather than on hidden representatives. If $q : \Sigma \to \Sigma/\Gamma$ is the quotient by boundary redundancy and $\overline{T}_i$ is the physical quotient update, a representative-level map T_i is only required to be a lift satisfying

$$q \circ T_i = \overline{T}_i \circ q.$$

Hence

$$q(T_i(\gamma \cdot s)) = q(T_i(s))$$

for gauge-equivalent inputs. Quotient descent is therefore structural, while strict representative-level covariance is only implementation bookkeeping. The burden is to prove that the accepted recovery-derived local moves satisfy the stated repair-completeness, support-local disjoint-commutation, nested-collar restriction-compatibility, and Petz-domain control clauses on the declared branch. The touched-overlap acceptance contract yields finite Lyapunov descent and derived termination for accepted moves, while the fixed-cutoff gluing package carries the parenthesization-invariant union-collar state used for the local diamond.

Proposition B.1 (Ancilla-stable UV underdetermination)**.** *Let a fixed-cutoff OPH realization be stabilized by finite ancillary factors K_P in a fixed product state, with observable patch algebras embedded as $\mathcal{A}(P) \otimes \mathbf{1}_{K_P}$ and repair dynamics acting trivially on the ancillas. Then observable expectations on the physical subalgebras, overlap data, the local-Gibbs branch, the collar conditional mutual information $I(A : D \mid B)$, the Fawzi–Renner remainder, the collar Markov modulus, and the quotient normal form are unchanged. Thus the fixed-cutoff theorem package determines the UV branch only modulo such ancillary stabilization together with gauge or implementation hiding, not a unique microscopic presentation.*

Proof. Product ancillas leave physical observables unchanged, cancel additively inside conditional mutual information, and are inert under the repair maps. Hence every invariant listed above is unchanged. □

B.2 Derived Boundary Data and Ordinary EC

Proposition B.2 (Derived boundary gluing datum)**.** *Choose a finite regulator chart for the patches meeting along a connected cut Σ. Because the local overlap algebras are finite-dimensional matrix algebras, any overlap-consistent recharting is an inner automorphism and is implemented by a unitary on the cut Hilbert space. The compact closure of the subgroup generated by these recharting unitaries is a compact boundary redundancy group K_Σ. If triple-overlap defects are central, the projective composition law lifts to a compact central extension $\widehat{K_\Sigma}$; on the ordinary branch one simply sets $\widehat{K_\Sigma} = K_\Sigma$. A genuinely noncentral 2-group defect is the only obstruction to reducing the overlap transition system to an ordinary compact group action.*

Theorem B.3 (Derived EC decomposition)**.** *Under the fixed-cutoff regulator realization above, and on the ordinary or central-defect branch, the collar Hilbert space is*

$$\mathcal{H}_{B_\delta} = (\tilde{\mathcal{H}}_{B_L} \otimes \tilde{\mathcal{H}}_{B_R})^{\widehat{K}_\Sigma} \cong \bigoplus_\alpha \left(\mathcal{H}_{b_L^\alpha} \otimes \mathcal{H}_{b_R^\alpha}\right),$$

and the center of the collar algebra is generated by the block projectors:

$$Z(\mathcal{A}(B_\delta)) = \bigoplus_\alpha \mathbb{C} \cdot \mathbf{1}_\alpha.$$

The right half-collar carries the contragredient representation because it sees inverse transport across the same cut.

Remark B.4. This is the finite-patch-net origin of the collar center used by the later Markov, record, and observer packages. Exact Markovity is an additional state hypothesis; EC provides the kinematic block structure.

B.3 Higher-Gauge Replacement on the Genuinely Noncentral Branch

Proposition B.5 (Derived higher-gauge cut datum)**.** *On the genuinely noncentral branch, weak overlap gluing on a connected cut Σ is encoded by a compact crossed module*

$$\mathbb{K}_\Sigma = (H_\Sigma \xrightarrow{\partial_\Sigma} G_\Sigma, \triangleright)$$

with defect class

$$q_\Sigma \in \check{H}^2(N_\Sigma, H_\Sigma \to G_\Sigma),$$

and compact higher-gauge change system

$$\mathcal{T}_\Sigma = C^1(N_\Sigma, H_\Sigma) \rtimes C^0(N_\Sigma, G_\Sigma).$$

Theorem B.6 (Higher-gauge EC decomposition and defect transport)**.** *On the genuinely noncentral branch,*

$$\mathcal{H}^{2g}_{B_\delta} = (\tilde{\mathcal{H}}_{B_L} \otimes \tilde{\mathcal{H}}_{B_R})^{\mathcal{T}_\Sigma} \cong \bigoplus_\lambda (\mathcal{H}_{b_L^\lambda} \otimes \mathcal{H}_{b_R^\lambda}),$$

and

$$Z(\mathcal{A}_{2g}(B_\delta)) = \bigoplus_\lambda \mathbb{C} \cdot \mathbf{1}_\lambda.$$

Moreover the full crossed-module orbit q_Σ is invariant under local rechartings and classifies fixed-cutoff genuinely noncentral gluing data. The higher associator is removable iff q_Σ lies in the image of

$$\check{H}^1(N_\Sigma, G_\Sigma) longrightarrow \check{H}^2(N_\Sigma, H_\Sigma \to G_\Sigma), \qquad [g] \longmapsto [(g, 1)].$$

That map need not be injective. Strict endpoint-only ordinary transport additionally requires at least one allowed strict representative with trivial represented loop holonomy.

Corollary B.7 (Exact Markov plus split alignment adds the state factorization)**.** *On either the ordinary/central branch of the previous theorem or the genuinely noncentral higher-gauge branch, if in addition*

$$I_\omega(A_\delta : D_\delta \mid B_\delta) = 0$$

and *the state satisfies the Markov-split alignment hypothesis (its HJPW decomposition of $\mathcal{H}_{B_\delta}$ can be chosen to be the EC decomposition itself; see the main-text Section 2.3 and the compact paper's alignment definition), or one passes to the explicitly stated idealized recoverability limit that supplies both conditions, then*

$$\rho_{A_\delta B_\delta D_\delta} = \bigoplus_\alpha p_\alpha \left(\rho_{A_\delta b_L^\alpha} \otimes \rho_{b_R^\alpha D_\delta} \right).$$

EC therefore gives the kinematic block decomposition, while exact Markovity plus split alignment is the extra state input that gives the EC-aligned HJPW normal form. Exact Markovity alone yields the normal form only over a state-dependent HJPW split, which a Bell-pair example (main text, Section 2.3) shows can be transposed relative to the EC factors.

Proposition B.8 (Parenthesization-invariant union-collar gluing)**.** *Fix a finite union collar U built from two overlapping local repair collars on the declared fixed-cutoff branch. On the ordinary or central-defect branch, the physical glued state on U determined by the exterior marginals and sector data is independent of how the local gluings are parenthesized, up to the boundary-redundancy action. On the genuinely noncentral branch, the same statement holds up to the crossed-module change system $\mathcal{T}_\Sigma$. Hence the physical quotient-local glued state on U is well defined independently of parenthesization.*

Proof. On the ordinary or central-defect branch, the preceding edge-center decomposition writes the collar algebra as a direct sum of sector blocks generated by central projectors. Reparenthesizing two overlapping local gluings changes only the representative by the central boundary action and cannot move the state between those block projectors, so the quotient-local glued state is unchanged. On the genuinely noncentral branch, the failure of strict composition is recorded by the crossed-module associator, and the higher-gauge theorem above identifies rechartings exactly with the $\mathcal{T}_\Sigma$-coboundary action. Reparenthesization therefore changes only the representative inside one $\mathcal{T}_\Sigma$-orbit. In either case the physical quotient state is parenthesization-independent. □

Remark B.9. Approximate recoverability gives controlled deviations from this normal form; it is not implied by EC alone. This is the fixed-cutoff topological package behind the consensus paper's quotient and record-language surface.

Corollary B.10 (Physical observables are invariant on one quotient-local glued state)**.** *Let U be a finite union collar on the declared fixed-cutoff branch, and let ω_U, ω'_U be two microscopic representatives of the same quotient-local glued state from Proposition B.8. Then every physical observable X on the collar fixed-point / quotient-local algebra has the same expectation in both representatives:*

$$\mathrm{Tr}(X\omega_U) = \mathrm{Tr}(X\omega'_U).$$

In particular the same holds for the central sector projectors and for any observer-accessible record observable generated from them on that same declared surface.

Proof. On the ordinary or central-defect branch, Proposition B.8 says the two representatives differ only by the boundary-redundancy action inside one fixed sector block. The fixed-point collar algebra and its central block projectors are invariant under that action, so their expectation values agree. On the genuinely noncentral branch, the same proposition says the two representatives differ only inside one $\mathcal{T}_\Sigma$-orbit, and the quotient-local physical algebra $\mathcal{A}_{\mathrm{phys}}(U)$ of Definition 5.16 is defined precisely on that orbit space. Therefore the induced physical state and all expectations of physical observables agree there as well. □

C Assumption-Dependent Distributed-Systems and QECC Extensions of the Consensus Formalism

Support labels.

[**Established**] Follows from cited prior work or a complete argument given here.

[**Assumption-dependent**] True under additional assumptions not derived from OPH first principles.

[**Conjecture**] A plausible open direction, not a settled result.

B.1 Theorem 1: QBFT Safety Bound

Definition C.1 (QBFT-style protocol)**.** *A consensus protocol is* QBFT-style *in this analysis if it satisfies the following three structural properties. The safety proof of Theorem C.2 uses all three; the theorem does not hold for protocols lacking any of them without a compensating change to the argument.*

(P1) ***One-vote-per-view.*** *Each nonfaulty node casts at most one vote per view number. A node that has voted in view v ignores any later request to vote in view v.*

(P2) ***Certificate semantics.*** *A decision requires a valid quorum certificate: in the classical exact-size case used below, $q = 2f + 1$ distinct, unforgeable, authenticated votes for the same value in the same view. For larger validator sets at fixed fault budget, the quorum threshold must be scaled so that the overlap condition in (A6) remains true.*

(P3) ***DLS-style view-change.*** *If no certificate is produced within a timeout, every nonfaulty node increments the view number by one and a new leader is selected by a fixed deterministic rule. At GST, timeouts fire correctly and the view-change terminates in bounded rounds.*

The Istanbul BFT / QBFT protocol family [7, 8] satisfies (P1)–(P3) and is the intended instance.

Assumptions A1–A6.

(A1) **Partial synchrony (DLS).** Fixed but initially unknown bounds Δ (message delay) and Φ (processing rates). *Safety* holds without extra timing assumptions; *liveness* holds after the Global Stabilisation Time (GST).

(B2) **Byzantine fault model.** At most f observers behave arbitrarily; the remaining $n - f$ are nonfaulty.

(C3) **Classical exact sizing for this fixed-quorum theorem.** $n = 3f + 1$, so that $q = 2f + 1$ certificates have a nonfaulty overlap witness. The usual resilience condition $n \geq 3f + 1$ is necessary for the fault model, but when $n > 3f + 1$ a fixed $q = 2f + 1$ quorum is insufficient for the overlap used in the safety proof; one must either impose (A6) directly or choose a threshold q with $2q \geq n + f + 1$.

(D4) **Strong quorum connectivity.** Every quorum Q with $|Q| = q$ is strongly connected within G: for any $u, v \in Q$ there is a directed path in G contained entirely in Q. This is strictly stronger than requiring the overlap graph of quorums to be connected, and is needed to propagate signed votes within a quorum.

(E5) **Message authentication.** All messages carry unforgeable digital signatures.

(F6) **OPH quorum overlap.** Any two quorums Q_a, Q_b of size q satisfy $|Q_a \cap Q_b| \geq f + 1$. For $q = 2f + 1$ this is guaranteed by the exact sizing $n = 3f + 1$ in (A3); for general n it is the separate threshold condition $2q \geq n + f + 1$, not a consequence of $n \geq 3f + 1$ alone.

D. Matscheko's Lean audit of this appendix checks the finite quorum-overlap core and records the same boundary caveat: at fixed $q = 2f + 1$, the overlap step is exact-size $n = 3f + 1$ logic unless (A6) is imposed separately.

Theorem C.2 (QBFT Safety Bound [Same-view case established under A1–A6; cross-view safety conditional on a (P4) locking rule]). *Under assumptions (A1)–(A6), any consensus protocol satisfying (P1)–(P3) of Definition C.1 and run over the OPH observer graph satisfies:*

(i) ***Same-view safety.*** *No two nonfaulty observers finalise conflicting patch states in the same view.*

(ii) ***Liveness.*** *After GST, every nonfaulty observer finalises within $O(f \cdot \Delta)$ wall-clock time.*

(iii) ***Optimality.*** *The bound $f < n/3$ is tight.*

Cross-view boundary. Clause (i) is stated for same-view finalisation because the argument below covers exactly that case. Cross-view safety in the IBFT/QBFT protocol family uses an additional prepared-certificate locking property — an explicit (P4) hypothesis under which a nonfaulty node votes in a later view only for a value carried by the highest prepared certificate it has received. (P1)–(P3) alone do not exclude finalisation of conflicting values across views; a cross-view extension of this theorem requires (P4) and its accompanying view-change justification argument. On its stated assumptions, this finalisation theorem is a finite carrier of stack row C1 of `CONSISTENCY_STACK.md`: record permanence—no two nonfaulty observers finalising conflicting patch states—enters as a consistency requirement on the observer net, not as an added postulate.

Proof sketch. Safety. Suppose O_a and O_b finalise $s_a \neq s_b$ in the same view. By (P2), each required a certificate of q votes: sets Q_a, Q_b. By (A6), $|Q_a \cap Q_b| \geq f + 1$. In the classical exact-size case this is the inclusion-exclusion calculation $|Q_a \cap Q_b| \geq (2f+1) + (2f+1) - (3f+1) = f + 1$. By (A2), at most f are Byzantine, so $Q_a \cap Q_b$ contains a nonfaulty O^*. By (A4), O^*'s signed vote is path-reachable within both quorums. By (P1), O^* voted for at most one value. Contradiction.

Liveness and Optimality follow from [6] (Thm. 4.4) and [5], cited directly.

Note on FLP. Fischer, Lynch, Paterson [4] is an impossibility result for fully asynchronous systems; it does not bear on achievability under partial synchrony (A1). □

B.2 Theorem 2: Convergence of the OPH Repair Map

Definition C.3 (OPH Repair Map: Petz form)**.** *Let $\sigma \in \mathcal{D}(\mathcal{H})$ be a full-rank reference state and $\mathcal{N} : \mathcal{B}(\mathcal{H}) \to \mathcal{B}(\mathcal{K})$ a quantum channel. The* OPH repair map *is*

$$\mathcal{R}_{\sigma,\mathcal{N}}(\rho) := \sigma^{1/2}\, \mathcal{N}^{\dagger}(\mathcal{N}(\sigma)^{-1/2}\, \rho\, \mathcal{N}(\sigma)^{-1/2})\, \sigma^{1/2},$$

where $\mathcal{N}^{\dagger}$ is the adjoint channel and inverses are taken on $\operatorname{supp}(\mathcal{N}(\sigma))$*.*

Remark C.4 (Petz map vs. trace-distance projection). The closest-point trace-distance projection $\mathcal{P}_{\mathcal{S}}(\rho) := \arg\min_{\tau \in \mathcal{S}} \frac{1}{2}\|\rho - \tau\|_1$ is a different object from the Petz map: it is defined by a variational problem in trace-norm geometry and is not CPTP in general. The two coincide only in very special cases not automatic in the OPH setting. All subsequent properties refer exclusively to Definition C.3.

Proposition C.5 (Petz map CPTP: domain-restricted statement [Established, subject to domain restriction])**.** *Let σ have full support on $\mathcal{H}$.*

(a) $\mathcal{R}_{\sigma,\mathcal{N}}$ is completely positive.

(b) $\mathcal{R}_{\sigma,\mathcal{N}}$ is trace-preserving on $\operatorname{supp}(\mathcal{N}(\sigma))$*, i.e., on inputs ρ for which $\mathcal{N}(\sigma)^{-1/2}\rho\mathcal{N}(\sigma)^{-1/2}$ is well-defined.*

(c) If additionally $\mathcal{N}(\sigma)$ has full rank on $\mathcal{K}$, then $\mathcal{R}_{\sigma,\mathcal{N}}$ is CPTP on all of $\mathcal{B}(\mathcal{K})$.

If $\mathcal{N}(\sigma)$ is not *full rank on $\mathcal{K}$, then either (i) the domain must be restricted to* $\operatorname{supp}(\mathcal{N}(\sigma))$*, or (ii) pseudoinverses must replace the inverses (generalised Petz map; cf. [11]), or (iii) a regularisation $\mathcal{N}(\sigma) \mapsto \mathcal{N}(\sigma) + \varepsilon\mathbf{1}$ must be introduced. Note that full-rank σ does* not *prevent $\mathcal{N}(\sigma)$ from being rank-deficient: the channel may map the support of σ into a strict subspace of $\mathcal{K}$. In the OPH setting, whether $\mathcal{N}(\sigma)$ is full rank depends on the specific overlap channel and must be verified for the chosen analytic channel model. The finite OPH repair theorem instead uses the declared Lyapunov descent law on a finite patch net.*

Proof. Complete positivity follows from composing three CP operations:

(i) sandwiching by $\mathcal{N}(\sigma)^{-1/2}(\cdot)\mathcal{N}(\sigma)^{-1/2}$ on $\operatorname{supp}(\mathcal{N}(\sigma))$;

(ii) $\mathcal{N}^{\dagger}$;

(iii) sandwiching by $\sigma^{1/2}(\cdot)\sigma^{1/2}$.

Trace preservation in the full-rank case: Petz [9]; Fagnola–Umanità [10]. □

Proposition C.6 (Analytic contraction certificate [Assumption-dependent]). *Suppose a declared quotient repair map $T : Q \to Q$ on a metric physical quotient (Q, d_Q) is strictly contractive with coefficient $\lambda \in (0, 1)$:*

$$d_Q(Tx, Ty) \le \lambda d_Q(x, y).$$

Then T has at most one fixed point. If a fixed point $x_\star$ exists, the ideal iterates obey $d_Q(T^t x, x_\star) \le \lambda^t d_Q(x, x_\star)$. This is an analytic contraction condition. It is not required for the finite OPH normal-form theorem, where termination follows from strict Lyapunov descent of accepted repairs.

Theorem C.7 (Noisy approximate repair stability [Contraction branch]). *Assume the contraction certificate of Proposition C.6, and let $\widetilde{T} : Q \to Q$ be an implemented noisy repair map satisfying*

$$d_Q(\widetilde{T}x, Tx) \le \varepsilon \qquad \text{for all } x \in Q.$$

If $x_\star$ is the ideal fixed point of T, then

$$d_Q(\widetilde{T}^t x, x_\star) \le \lambda^t d_Q(x, x_\star) + \frac{\varepsilon}{1 - \lambda}.$$

Thus the noisy branch converges only to a controlled error ball, and only after the contraction certificate and uniform implementation-error bound are supplied.

Proof. The recursion

$$d_Q(\widetilde{T}x, x_\star) \le d_Q(\widetilde{T}x, Tx) + d_Q(Tx, Tx_\star) \le \varepsilon + \lambda d_Q(x, x_\star)$$

iterates to the displayed geometric-series bound. □

Proposition C.8 (Spectral-gap criterion [Model-dependent]). *Let $\mathcal{T}$ be the Markov, channel, or transfer operator induced by iterated OPH repair on a declared analytic realization. If $\mathcal{T}$ has stationary projection $\Pi_\star$ and constants $C < \infty$, $\delta > 0$ such that on the nonstationary subspace*

$$\|\mathcal{T}^t - \Pi_\star\| \le Ce^{-\delta t},$$

then the corresponding analytic channel model has exponential convergence. The finite OPH repair package supplies termination by Lyapunov descent on its declared finite state space; a spectral gap is a separate quantitative mixing condition for this BFT/QECC-style extension.

Theorem C.9 (Exponential Convergence [Under Proposition C.8]). *Under Proposition C.8, for any initial analytic state ρ,*

$$\|\mathcal{T}^t \rho - \Pi_\star \rho\| \le C\, e^{-\delta t} \|\rho - \Pi_\star \rho\|.$$

This theorem belongs only to the spectral-gap branch. It is not a consequence of a bare finite overlap graph or of finite Lyapunov descent alone.

B.3 Theorem 3: QECC Correspondence

Notation. $N = \dim(\mathcal{H}) = 2^n$ for n physical qubits. Standard notation: $[[n, k, d]]$ stabilizer code; $K = 2^k$; quantum Singleton bound: $k \le n - 2(d - 1)$.

Theorem C.10 (No free min-cut theorem for bare overlap graphs [Established]). *Let $G = (V, E)$ be any connected graph with $|V| \ge 2$. The graph G alone does not determine the Hamming distance of the consistency set C of a finite overlap net on G. In particular, the same graph can realize a binary constraint code of distance 1 or a binary repetition code of distance $|V|$.*

Proof. Set $S_i = \{0,1\}$ for every vertex. For the repetition realization, choose $I_e = \{0,1\}$ and let both endpoint readouts be the identity. Then every edge imposes $x_i = x_j$, so the only global codewords are $00\cdots0$ and $11\cdots1$, whose Hamming distance is $|V|$.

For the trivial-overlap realization, keep the same vertex state spaces but let every endpoint readout be the constant map to 0. Then every binary assignment is globally consistent, so the minimum Hamming distance among distinct codewords is 1. The graph is unchanged. Therefore distance is a property of the code realization–state spaces, readout maps, logical dictionary, metric, and error model–not of the bare overlap graph. □

Definition C.11 (Topological-code realization certificate)**.** *An OPH overlap network may be treated as a QECC/topological code only after supplying a tuple*

$$\mathsf{TCert} = (K, \mathcal{H}_{\mathrm{phys}}, \mathcal{H}_{\mathrm{code}}, \partial_2, \partial_1, S_X, S_Z, \mathcal{L}_X, \mathcal{L}_Z, \mathcal{E}, \mathcal{R}),$$

where $C_2 \xrightarrow{\partial_2} C_1 \xrightarrow{\partial_1} C_0$ is a chain complex over $\mathbb{F}_2$, the physical carriers live in $\mathcal{H}_{\mathrm{phys}}$, the protected subspace is $\mathcal{H}_{\mathrm{code}}$, S_X, S_Z are stabilizer or gauge checks, $\mathcal{L}_X, \mathcal{L}_Z$ are logical-operator classes, $\mathcal{E}$ is a declared error family, and $\mathcal{R}$ is a recovery map or recovery family.

Theorem C.12 (Certified topological-code distance and min-cut [Assumption-dependent])**.** *Suppose a certificate* TCert *of Definition C.11 is supplied, with logical classes identified as*

$$\mathcal{L}_X \simeq H_1(K;\mathbb{F}_2), \qquad \mathcal{L}_Z \simeq H^1(K;\mathbb{F}_2),$$

and with boundary conditions excluding lower-weight trivial representatives. Then the certified distance is

$$d = \min\left\{ \min_{\ell \in \mathcal{L}_X \setminus 0} |\ell|, \min_{\ell^\star \in \mathcal{L}_Z \setminus 0} |\ell^\star| \right\}.$$

Only in geometries where this homological systole equals the relevant graph min-cut may one write $d = \mathrm{mincut}(G_{\mathrm{OPH}})$ [13, 14].

Proof. This is the standard stabilizer/topological-code distance statement once the chain complex, checks, logical representatives, and boundary conditions are declared. The min-cut equality is an additional geometric identification of that homological minimum with a graph cut. By Theorem C.10, it cannot be inferred from the bare graph. □

Conjecture C.13 (Communication complexity [Conjecture])**.** *The OPH consensus-repair protocol, realised as a quantum communication task, has per-round complexity $O(n \cdot \mathrm{poly}(d))$ for a chosen communication encoding (cf. [15]). The fixed finite repair theorem gives termination after a supplied descent law; it does not by itself fix a quantum communication complexity class for every implementation.*

Theorem C.14 (QECC resilience under a supplied code certificate [Assumption-dependent])**.** *Assume a genuine code subspace $\mathcal{H}_{\mathrm{code}} \subseteq \mathcal{H}_{\mathrm{phys}}$ with projector Π, and let $\mathcal{E}_t$ be the declared set of errors supported on fewer than t corrupted patches or physical carriers. If*

$$\Pi E_a^\dagger E_b \Pi = \alpha_{ab} \Pi \qquad \forall E_a, E_b \in \mathcal{E}_t,$$

then there exists a recovery channel correcting all errors in $\mathcal{E}_t$ [12]. If the supplied certificate also gives distance d, then all errors of weight $t < d/2$ are correctable. No such resilience statement follows from the bare overlap graph.

Corollary C.15 (QECC extension inventory)**.** *Under Definition C.11, Theorem C.12, and Theorem C.14, the OPH BFT/QECC extension carries the following claim split:*

(i) [Assumption-dependent] Code distance is the certified homological minimum; it equals $\mathrm{mincut}(G_{\mathrm{OPH}})$ *only under the additional systole/min-cut identification.*

(ii) [Established] The Knill–Laflamme QECC theorem supplies recovery once the projector and error family satisfy the displayed condition.

(iii) [Conjecture] Per-round communication complexity is $O(n \cdot \mathrm{poly}(d))$*.*

B.4 Theorem 4: Asynchronous Convergence

Why fairness alone does not give a probability-1, spectral, or wall-clock statement. Standard strong fairness guarantees that every enabled action fires infinitely often along any fair schedule; it does not impose a probability space on schedules, a transfer-operator spectral gap, or a message-delay bound. A convergence statement of the form "converges with probability 1" requires a measure on schedules. Exponential convergence requires the spectral-gap certificate of Theorem C.9. Bounded wall-clock liveness requires partial synchrony and quorum assumptions. The FLP impossibility result [4] confirms that fairness is insufficient for bounded-time consensus in a fully asynchronous system.

Additional assumptions for a quantitative bound.

(B1) Finite known bound Δ on message delay after GST.

(B2) Finite bound Φ on processing rates.

(B3) $f < n/3$.

Theorem C.16 (Eventual finite repair termination [Finite-descent branch])**.** *For a finite OPH patch net with a strict Lyapunov-decreasing accepted repair relation, every maximal repair run terminates after finitely many accepted repairs. The step count is bounded by the finite value-set bound of Proposition 3.19. If the local-diamond and repair-completeness clauses also hold, Newman's lemma upgrades this termination statement to the unique schedule-independent quotient normal form of Theorem 3.22.*

This theorem is finite descent convergence in repair steps. It is not trace-norm convergence of a Petz channel, not probability-one convergence over random schedules, not exponential convergence, and not a wall-clock liveness theorem.

Theorem C.17 (Quantitative Convergence [Assuming (B1)–(B3)])**.** *In a partially synchronous OPH observer network satisfying (B1)–(B3), after GST every nonfaulty observer reaches consensus within* $T = O(f \cdot \Delta)$ *wall-clock time (by applying the DLS framework [6], Thm. 4.4, to the OPH repair protocol; requires (B1) and (B2) explicitly and does not follow from fairness alone).*

B.5 Extension Boundaries

- A bare OPH overlap graph is a finite constraint-code presentation only. Its graph does not determine code distance, correctable error weight, or min-cut resilience.

- Analytic spectral-gap and full-rank estimates for a chosen stochastic or channel-level BFT/QECC realization are model-specific refinements. They are separate from the finite OPH repair theorem, where the accepted repair law supplies Lyapunov descent directly.

- Long-run noisy approximate consensus is available only on the fair-block contraction branch of Theorem 11.1: the chosen implementation must certify fair blocks, expected contraction toward the exact quotient normal-form set, and controlled within-block excursions. Fairness alone does not supply that certificate.

- Topological-code distance equals graph min-cut only after a concrete chain complex, boundary condition, logical-operator dictionary, error family, and systole/min-cut identification are supplied for the chosen code realization.

- Communication-complexity bounds require a concrete quantum communication encoding and implementation cost model. They are not consequences of finite normal-form termination alone.

- The core OPH consensus paper supplies observable-level confluence and refinement-limit normal-form/holonomy classes on their declared theorem surfaces; the BFT/QECC statements above are separate protocol-style extensions.

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